Vel Tech High Tech

Dr.Rangarajan Dr.Sakunthala Engineering College

An Autonomous Institution

(AN ISO 9001-2008 CERTIFIED INSTITUTION)

(Approved by AICTE, New Delhi & Affiliated to Anna University)

No.60, Avadi – Vel Tech Road, Chennai-600 062.

21HC67P INTERNET PROGRAMMING LABORATORY

NAME :

REGISTER NO :

ROLL NO :

BRANCH : B.E-Computer Science and Engineering

YEAR : III

SEMESTER : VI

NAME : YUVARAJ G REG NO : 113023104181

Vel Tech High Tech

Dr.Rangarajan Dr.Sakunthala Engineering College

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No.60, Avadi – Vel Tech Road, Chennai-600 062.

BONAFIDE CERTIFICATE

Name.....

Year:……..Semester:……Branch: .....B.E–Computer Science and Engineering.....

University Register No: | | | | | | | | | | | | College Roll No: VH

Certified that this is the bonafide record of work done by the above student in the

21HC67P - Internet Programming Laboratory during the academic year 2025-2026.

Signature of Lab Incharge Signature of Head of the Department

Submitted for the University Practical Examination held on.....at VEL TECH HIGH TECH Dr. RANGARAJAN Dr. SAKUNTHALA ENGINEERING COLLEGE, No.60, AVADI – VEL TECH ROAD, AVADI, CHENNAI-600062.

Signature of Examiners

Internal Examiner:..... External Examiner:.....

Date:.....

NAME : YUVARAJ G REG NO : 113023104181

VEL TECH HIGH TECH DR.RANGARAJAN DR.SAKUNTHALA ENGINEERING COLLEGE

INSTITUTION VISION

Pursuit of excellence in technical education to create civic responsibility with competency.

INSTITUTION MISSION

1. To impart the attributes of global engineers to face industrial challenges with social relevance.
2. To indoctrinate as front runners through moral practices.
3. To attain the skills through life long learning.

DEPARTMENT VISION

To blend academic learning process and innovative ideas producing self-confident graduates with skills and knowledge to compete in the changing world.

DEPARTMENT MISSION

M1: To provide strong basic knowledge in Computer Science and Engineering.

M2: To motivate the students to function as a team from their innovative ideas in collaboration with the industries.

M3: To enable the students to lead and adapt to the growing environment.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1:

Embark upon successful professional practice in Computer Science and Engineering, displaying supportive and leadership roles.

PEO 2:

Engage in professional projects requiring teamwork and making valuable contributions to design, development, and production in the practice of Computer Science and Engineering or application areas.

PEO 3:

Equip to adapt and grow with changes in technology and globalization, and to pursue higher studies and research activities.

PEO 4:

Be capable of productive employment in the field of Computer Science and Engineering with competing technical expertise, good interpersonal skill.

PEO 5:

Utilize their broad educational experience, ethics, and professionalism to make a positive impact on their local and professional communities.

PROGRAMME SPECIFIC OUTCOMES (PSOs)

PSO's	PROGRAMME SPECIFIC OUTCOMES (PSOs)
PSO1	Designing Computer/Electronic based components which would serve social environment.
PSO2	Applying the current and gained knowledge and modern techniques not only in the Computers but in all related fields.

PROGRAM OUTCOMES:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineeringsciences.
3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilitiesrelevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineeringpractice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multi disciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi disciplinary environments.

VEL TECH HIGHTECH Dr. RANGARAJAN Dr. SAKUNTHALA ENGINEERING COLLEGE

COURSE OBJECTIVES:

- To be familiar with Web page design using HTML/XML and style sheets
- To be exposed to creation of user interfaces using Java frames and applets.
- To learn to create dynamic web pages using server side scripting.
- To learn to write Client Server applications.
- To be familiar with the PHP programming.
- To be exposed to creating applications with AJAX

COURSE OUTCOMES:

CO No	Course Outcomes	Knowledge Level
CO1	Construct Web pages using HTML/XML and style sheets.	K2
CO2	Build dynamic web pages with validation using Java Script objects and by applying different event handling mechanisms.	K3
CO3	Develop dynamic web pages using server side scripting.	K2
CO4	Use PHP programming to develop web applications.	K3
CO5	Construct web applications using AJAX and web services.	K2

COURSE OUTCOME AND PROGRAMME OUTCOME MAPPING:

CO	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2
CO 1	3	3	2	1	-	-	-	-	-	-	-	2	2
CO 2	3	3	2	1	-	-	-	-	-	-	-	2	2
CO 3	3	3	2	1	-	-	-	-	-	-	-	2	2
CO 4	3	3	2	1	-	-	-	-	-	-	-	2	3
CO 5	3	3	2	1	-	-	-	-	-	-	-	2	2

NAME : YUVARAJ G REG NO : 113023104181

CS8661 INTERNET PROGRAMMING LABORATORY L T P C
0 0 4 2

LIST OF EXPERIMENTS:

1. Create a web page with the following using HTML
a. To embed a map in a web page
b. To fix the hot spots in that map
c. Show all the related information when the hot spots are clicked.

2. Create a web page with the following.
a. Cascading style sheets.
b. Embedded style sheets.
c. Inline style sheets. Use our college information for the web pages.

3. Validate the Registration, user login, user profile and payment by credit card pages using JavaScript.

4. Write programs in Java using Servlets:
i. To invoke servlets from HTML forms
ii. Session tracking using hidden form fields and Session tracking for a hit count.

5. Write programs in Java to create three-tier applications using servlets for conducting online examination for displaying student mark list. Assume that student information is available in a database which has been stored in a database server.

6. Install TOMCAT web server. Convert the static web pages of programs into dynamic web pages using servlets (or JSP) and cookies. Hint: Users information (user id, password, credit card number) would be stored in web.xml. Each user should have a separate Shopping Cart.

7. Redo the previous task using JSP by converting the static web pages into dynamic web pages. Create a database with user information and books information. The books catalogue should be dynamically loaded from the database.

8. Create and save an XML document at the server, which contains 10 users Information. Write a Program, which takes user Id as an input and returns the User details by taking the user information from the XML document.

9. i. Validate the form using PHP regular expression.
ii. PHP stores a form data into database.

10. Write a web service for finding what people think by asking 500 people‘s opinion for any consumer product.

TOTAL: 60 PERIODS
SOFTWARE REQUIRED:
Dream Weaver or Equivalent, MySQL or Equivalent, Apache Server, WAMP/XAMPP.

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TABLE OF CONTENTS

EX.NO	DATE	TITLE OF THE PROGRAM	PAGE NO	MARK	SIGN
1		CREATING A WEBPAGE USING HTML			
2		CREATION OF WEB PAGE USING CASCADING STYLE SHEET			
3		FORM VALIDATION USING JAVASCRIPT			
4(a)		INVOKING SERVLETS FROM HTML FORM			
4b(i)		SESSION TRACKING USING HIDDEN FORM FIELDS			
4b(ii)		SESSION TRACKING A HIT COUNT			
5		THREE-TIER APPLICATIONS USING SERVLETS			
6		INSTALLING TOMCAT AND CONVERTING STATIC WEBPAGES TO DYNAMIC WEB PAGES			
7		CONVERTING STATIC WEB PAGES TO DYNAMIC WEB PAGES(USER INFORMATION AND BOOKS INFORMATION)			
8		RETRIEVING USER INFORMATION FROM XML DOCUMENT			
9		VALIDATING AND STORING A FORM DATA INTO DATABASE USING PHP			
10		A SURVEY USING WEB SERVICE FOR GETTING A FEEDBACK ON ANY CONSUMER PRODUCT			
Content Beyond Syllabus					
11		CREATING A HTML5 VIDEO PLAYER AND GETTING FEEDBACK			
12		TO ENCODE AND DECODE A STRING USING JAVA			

NAME : YUVARAJ G REG NO : 113023104181

Ex.No: 1	CREATING A WEBPAGE USING HTML
DATE:	

AIM:

ALGORITHM:

Step1 : Open a notepad.
Step 2: Write the code for imagemap.html.
Step 3: Enter a program that includes tags for <MAP> and other tags. Step 4: Insert Hyperlink using <A href>.
Step5: Save the file with .html extension.
Step6: Run the program in a web browser
Step7: Display Results.

PROGRAM:

Home.html

```
<html>
 <head>
 <title>Home - States of India!!!</title>
 </head>
 <body bgcolor="gold">
 <h1><u><center>Republic of India</center></u></h1>
 <p><h2>
India is the Seventh Largest country in the world by geographical area,
the second most populous country with over 1.2 billion people, and the populous democracy in
the world.</h2>
 </p>
<h1> Features</h1>
 <ul>
 <li><b><h2>Population</b> - 1,028,783,343(2001 census).</h2>
 <li><b><h2>Capital</b> - New Delhi</h2>
 <li><b><h2>Currency</b> - Indian Rupee</h2>
 <li><b><h2>Nation Sport</b> - Hockey</h2>
 </li>
 </ul>
  <center>
 <img align="left" size="8" src="IndiaMap.jpg" usemap="#india"/>
 <map name="india">
 <area shape="circle" coords="348,600,20" href="ANDHRAPRADESH.html">
 <area shape="circle" coords="289,760,10" href="KERALA.html">
 <area shape="circle" coords="266,620,35" href="KARNATAKA.html">
 <area shape="circle" coords="336,730,35" href="TAMILNADU.html">
 </map>
 </center>
 <h1><center> <b>To view details of states please click on the specified area in the map
!!!</b></center>
</h1>
</body>
</html>
```

Tamilnadu.html

```
<html>
<head>
<title>Tanil Nadu - India</title></head>
<body bgcolor="palegreen">
<h1><center>Tamil Nadu</center></h1>
<h3>It is one of the 28 states of India. Its capital and largest city is Chennai. Tamil Nadu lies in
the southernmost part of the Indian Peninsula and is bordered by the States of puducherry, Kerala,
Karnataka, Andhra Pradesh.</h3>
<h3>
```

```
<ul>
<li>Districts<i>            -
32</i></li><li>Capital
City<i> - Chennai</i>
</li><li>Largest City<i> - Chennai</i>
</li><li>Governor<i> - Banwarilal Purohit</i>
</li><li>Chief Minister<i> - Edappadi K. Palaniswami</i>
</li><li>Population<i> - 72,138,958</i>
</li><li>Tourist spots<i> - Mamallapuran, Ooty, Kodaikanal, Marina,Mudurai Meenakshi
Amman Temple, Thanjavur etc.,</i>
</li></ul>
<a href="Home.html">back</a>
</h3>
</body>
</html>
```

Andhrapradesh.html

```
<html>
<head>
<title>Andhra Pradesh - India</title>
</head>
 <body bgcolor="tan">
 <h1><center>Andhra Pradesh</center></h1>
 <h3>A.P, is a state situated on the southeastern coast of India. It is India's fourth largest state by
area and fifth largest by population.</h3>
 <h3>
 <ul>
 <li>Districts<i> - 23</i>
 </li><li>Capital City<i> - Hyderabad</i>
 </li><li>Largest City<i> - Hyderabad</i>
 </li><li>Governor<i> - Biswabhusan Harichandan </i>
 </li><li>Chief Minister<i> - Y. S. Jaganmohan Reddy </i>
</li><li>Population<i> - 78,323,330</i>
</li><li>Tourist spots<i> - Tirumala Tirupati, Charminar, Golconda Fort, Chandragiri,
Chowmahalla Place, Falaknuma Palace etc.,</i>
 </li></ul>
 <a href="Home.html">back</a>
</h3>
</body>
</html>
```

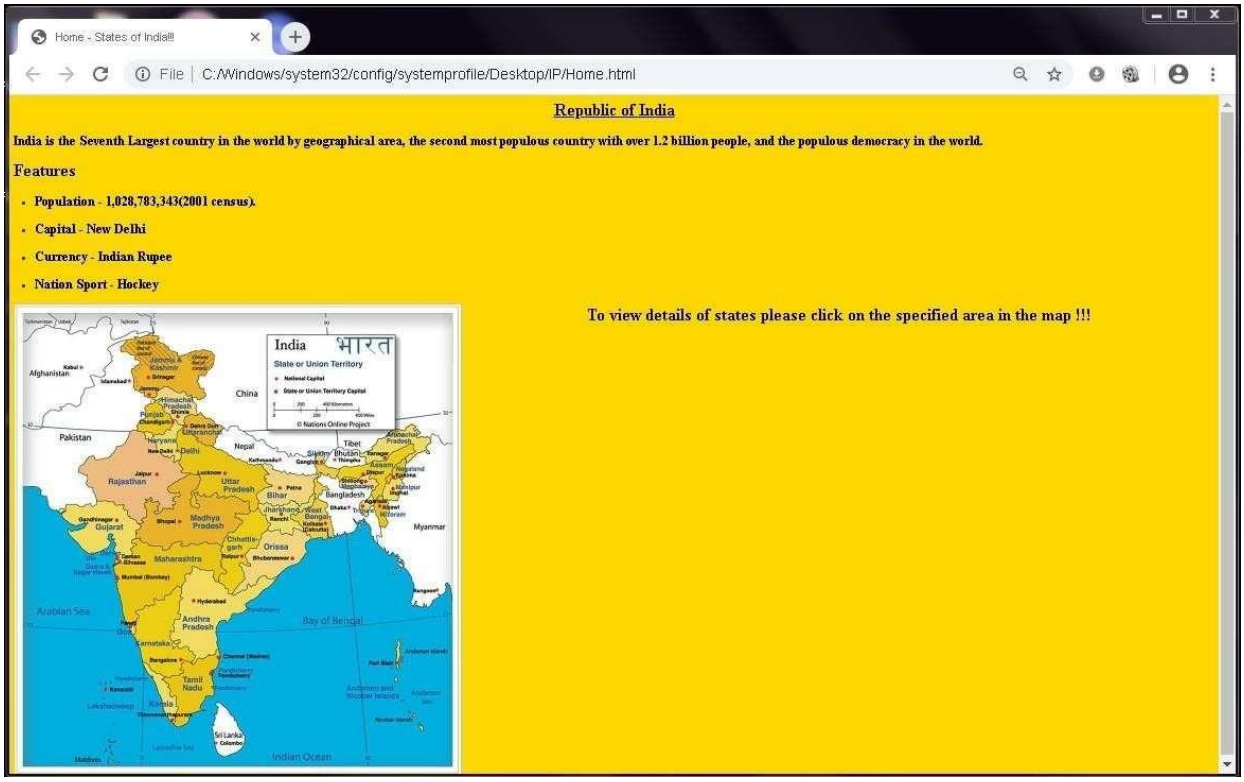
Kerala.html

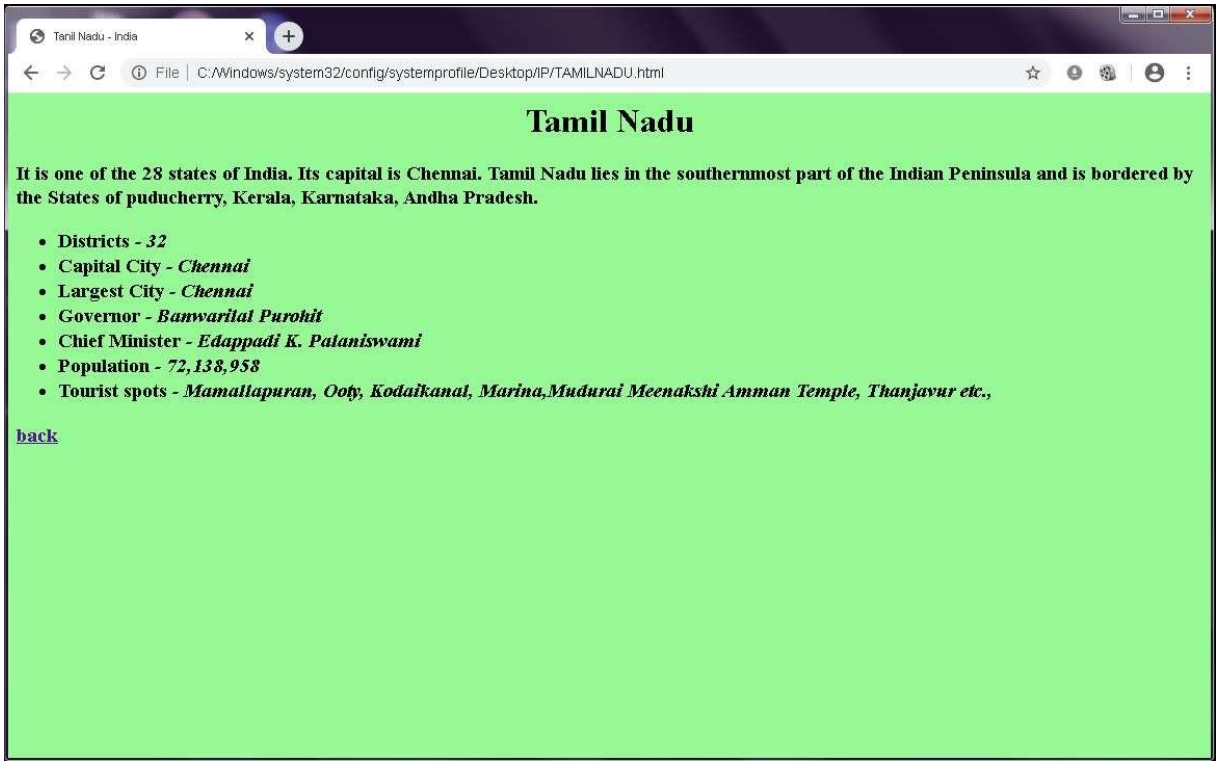
```
<html>
<head>
<title>Kerala - India</title>
</head>
<body bgcolor="indianred">
<h1><center>Kerala</center></h1>
<h3>
<ul>
<li>Districts<i> - 14</i>
</li><li>Capital City<i> - Thiruvanandapuram</i>

/li><li>Largest City<i> - Thiruvanandapuram</i>
</li><li>Governor<i> - Arif Mohammad Khan</i>
</li><li>Chief Minister<i> - Pinarayi Vijayan</i>
</li><li>Population<i> - 33,387,677</i>
</li><li>Tourist spots<i> - Edakkal Caves, Palayur, Kovalam Beach,Munnar, Kochi,Alapuzha
etc.,</i>
</li></ul>
</h3>
<a href="Home.html">Back</a>
</body>
</html>
```

Karnataka.html

```
<html>
<head>
<title>Karnataka - India</title>
</head>
<body bgcolor="wheat">
<h1><center>Karnataka</center></h1>
<h3>
<ul>
<li>Districts<i> - 30</i>
</li><li>Capital City<i> - Bangalore</i>
</li><li>Largest City<i> - Bangalore</i>
</li><li>Governor<i>- Vajubhai Vala</i>
</li><li>Chief Minister<i> - B. S. Yediyurappa</i>
</li><li>Population<i> - 61,130,704</i>
</li><li>Tourist spots<i> - Gol Gumbaz, Mysore Palace, Keshava Temple etc.,</i>
</li></ul>
</h3>
<a href="Home.html">back</a>
</body>
</html>
```

OUTPUT:


Home - States of India!!!
File | C:/Windows/system32/config/systemprofile/Desktop/IP/Home.html
Republic of India
India is the Seventh Largest country in the world by geographical area, the second most populous country with over 1.2 billion people, and the populous democracy in the world.
Features
Population - 1,028,783,343(2001 census).
Capital - New Delhi
Currency - Indian Rupee
Nation Sport - Hockey
To view details of states please click on the specified area in the map !!!
India
भारत
State or Union Territory
© Nations Online Project


Tamil Nadu - India
File | C:/Windows/system32/config/systemprofile/Desktop/IP/TAMILNADU.html
Tamil Nadu
It is one of the 28 states of India. Its capital is Chennai. Tamil Nadu lies in the southernmost part of the Indian Peninsula and is bordered by the States of puducherry, Kerala, Karnataka, Andha Pradesh.
Districts - 32
Capital City - Chennai
Largest City - Chennai
Governor - Banwarilal Purohit
Chief Minister - Edappadi K. Palaniswami
Population - 72,138,958
Tourist spots - Mamallapuran, Ooty, Kodaikanal, Marina,Mudurai Meenakshi Amman Temple, Thanjavur etc.,
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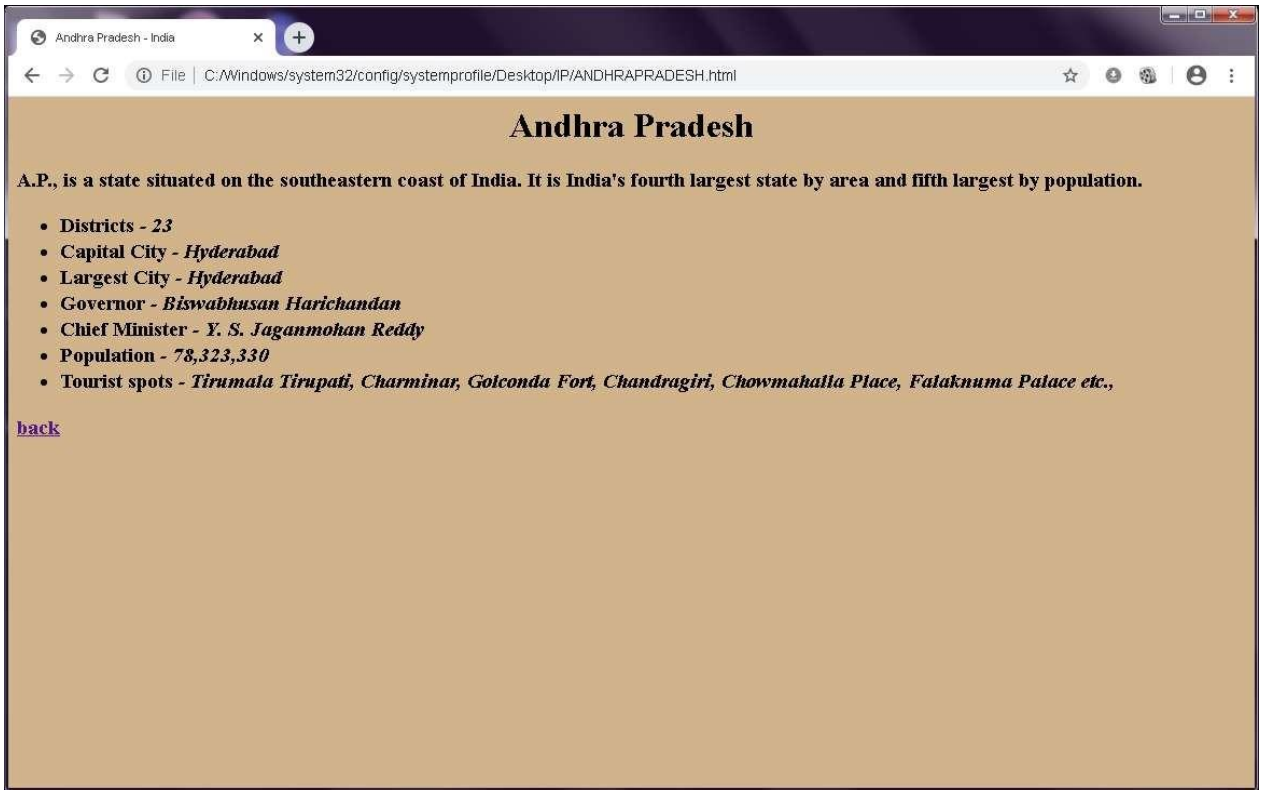
Andhra Pradesh - India
File | C:/Windows/system32/config/systemprofile/Desktop/IP/ANDHRAPRADESH.html
Andhra Pradesh
A.P., is a state situated on the southeastern coast of India. It is India's fourth largest state by area and fifth largest by population.
Districts - 23
Capital City - Hyderabad
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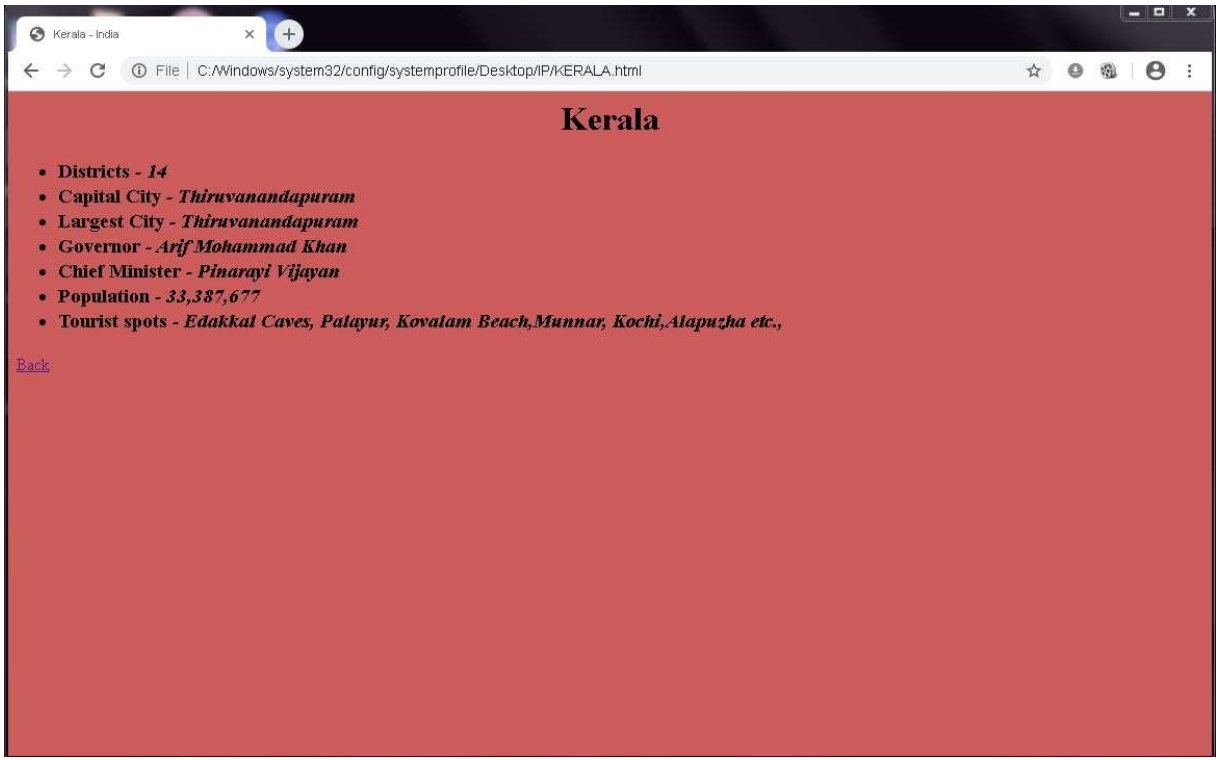
Population - 78,323,330
Tourist spots - Tirumala Tirupati, Charminar, Golconda Fort, Chandragiri, Chowmahalla Place, Falaknuma Palace etc.,
back


Kerala - India
File | C:/Windows/system32/config/systemprofile/Desktop/IP/KERALA.html
Kerala
Districts - 14
Capital City - Thiruvanandapuram
Largest City - Thiruvanandapuram
Governor - Arif Mohammad Khan
Chief Minister - Pinarayi Vijayan
Population - 33,387,677
Tourist spots - Edakkal Caves, Palayur, Kovalam Beach,Munnar, Kochi,Alapuzha etc.,
Back

Karnataka - India
File | C:/Windows/system32/config/systemprofile/Desktop/IP/KARNATAKA.html
Karnataka
Districts - 30
Capital City - Bangalore
Largest City - Bangalore
Governor- Vajubhai Vala
Chief Minister - B. S. Yediyurappa
Population - 61,130,704
Tourist spots - Gol Gumbaz, Mysore Palace, Keshava Temple etc.,
back

RESULT:

Ex.No: 2	CREATION OF WEB PAGE USING CASCADING STYLE SHEET
DATE:	

AIM:

ALGORITHM:

Inline CSS:

Step 1: Create a HTML program with <style> tag.
Step 2: Inside each html element specify the format required for that web page using<style> tag.
Step3 : Run the program with a web browser.

Internal CSS:

Step 1: Create a HTML program with <style> tag.
Step 2: Inside the <style> tag, specify the format required for that web page.
Step3 : Run the program with a web browser.

External CSS:

Step 1: Open a notepad, type the needed CSS in it and save it with .css extension.
Step2: Refer this .css file in the HTML using the tag <link>.
Step3: Run the program with a web browser.

PROGRAM:

InlineCSS.html

```
<html xmlns="http://www.w3.org/1999/xhtml"><head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <title>Inline CSS</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1 style="color:blue;text-align:center;">Example for External CSS</h1>
    <p style="color:red;">Vel Tech High Tech Dr.Rangarajan Dr.Sakunthala Engineering College was established in the year 2002 under R.S.Trust, founded by the Philanthropic couple: Col. Prof. Vel. Dr. R. Rangarajan, Founder President & Chairman and Dr. Sagunthala Rangarajan, Foundress Vice-Chairman respectively. Under the governance of AICTE & Anna University, this College has 11 branches of study with 8 UG Programmes and 3 PG Programmes</p>
  <h2 style="color:green;text-align:center;">Image Affected with styles</h2>
  <img src="college_logo.png"><br><br>
  <a href="college_logo.png" target="_blank">Download Image</a>
</body>
</html>
```

Internal CSS.html

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
  <title>Internal CSS</title>
  <style>
    body{
    background-color:#9F6;
    }
    h1{ color:#C39;
    text-align:left;
    text-transform:capitalize;
    text-decoration:underline;
    }
    P{
    font-size:20px;
    font-family:Verdana, Geneva, sans-serif;
    background-color:#FFF;
    color:#963;
    } h2{
    color:#F03;
    margin-left:10px;
    }
    img{
```

```
      border:5px          double #903;
      width:400px;
      height:300px;
      margin-left:10px;
      }
      a{
      margin-left:10px;
      } a:hover{
      color:#009; font-
      size:18px;
      font-weight:700;
      }
    </style>
  </head>
  <body>
    <h1>Example for Internal CSS</h1>
    <p> Vel Tech High Tech Dr.Rangarajan Dr.Sakunthala Engineering College was established in the year 2002 under R.S.Trust, founded by the Philanthropic couple: Col. Prof. Vel. Dr. R. Rangarajan, Founder President & Chairman and Dr. Sagunthala Rangarajan, Foundress Vice-Chairman respectively. Under the governance of AICTE & Anna University, this College has 11 branches of study with 8 UG Programmes and 3 PG Programmes</p>
    <h2>Image Affected with styles</h2>
    <img src="college_logo.png"><br><br>
    <a href="college_logo.png" target="_blank">Download Image</a>
  </body>
  </html>
```

Style.css

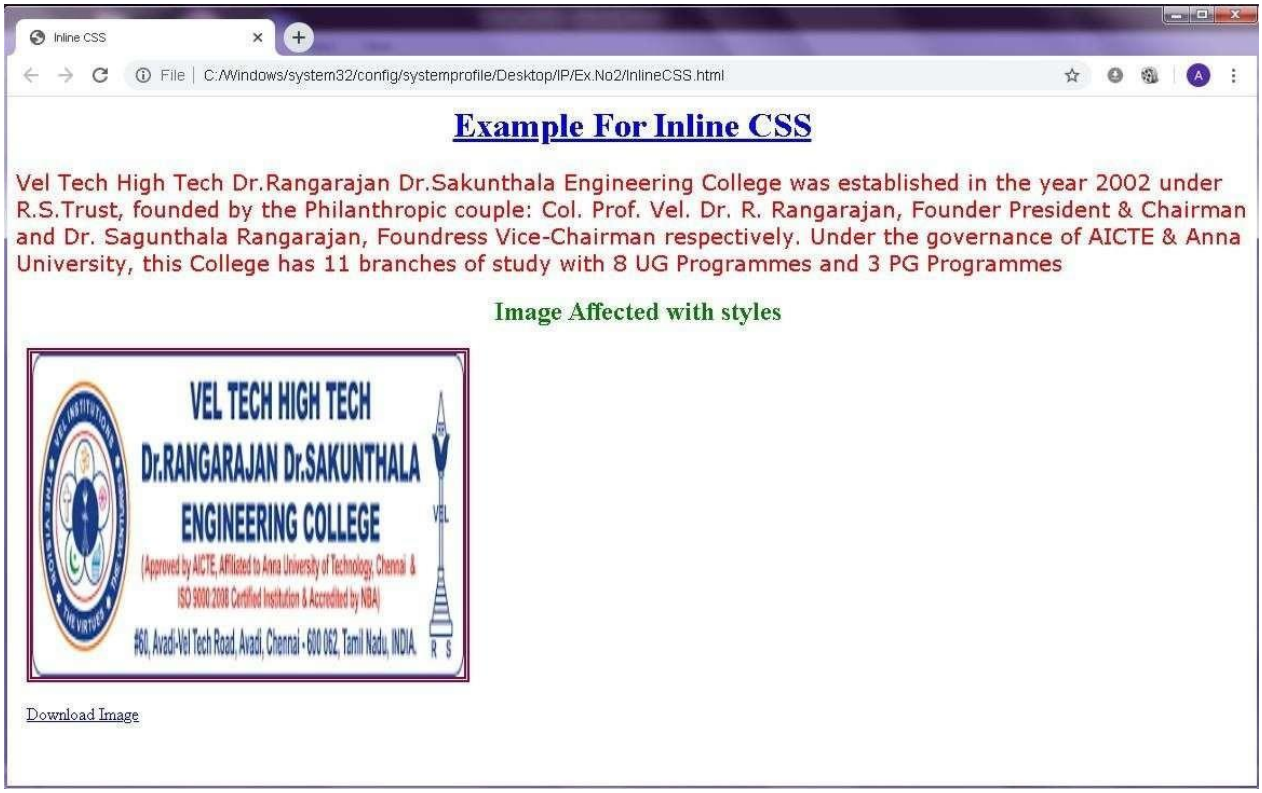
```
  <style>
      body{
      background-color:#9F6;
      }
      h1{ color:#C39;
      text-align:left;
      text-transform:capitalize;
      text-decoration:underline;
      }
      P{
      font-size:20px;
      font-family:Verdana, Geneva, sans-serif;
      background-color:#FFF;
      color:#963;
      } h2{
      color:#F03;
      margin-left:10px;
      }
```

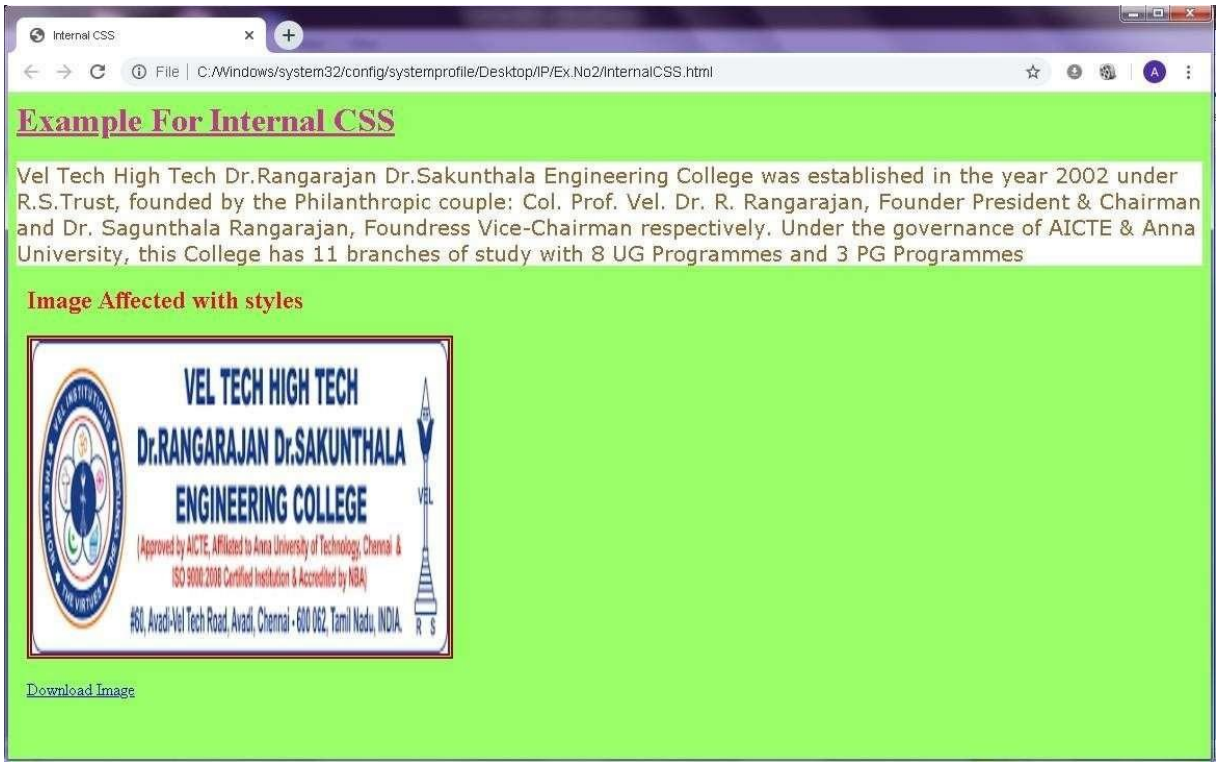
```
    img{
    border:5px          double #903;
    width:400px;
    height:300px;
    margin-left:10px;
    }
    a{
    margin-left:10px;
    } a:hover{
    color:#009; font-
    size:18px;
    font-weight:700;
    }
  </style>
```

External CSS.html

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
  <title>External CSS</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Example for External CSS</h1>
  <p>Vel Tech High Tech Dr.Rangarajan Dr.Sakunthala Engineering College was established in
the year 2002 under R.S.Trust, founded by the Philanthropic couple – Col. Prof. Vel. Dr. R.
Rangarajan, Founder President & Chairman and Dr. Sagunthala Rangarajan, Foundress
Vice-Chairman respectively. Under the governance of AICTE & Anna University, this
College has 11 branches of study with 8 UG Programmes and 3 PG Programmes</p>
  <h2>Image Affected with styles</h2>
  <img src="college_logo.png"><br /><br />
  <a href="college_logo.png" target="_blank">Download Image</a>
</body>
</html>
```

OUTPUT:


Inline CSS
File | C:/Windows/system32/config/systemprofile/Desktop/IP/Ex.No2/InlineCSS.html
Example For Inline CSS
Vel Tech High Tech Dr.Rangarajan Dr.Sakunthala Engineering College was established in the year 2002 under R.S.Trust, founded by the Philanthropic couple: Col. Prof. Vel. Dr. R. Rangarajan, Founder President & Chairman and Dr. Sagunthala Rangarajan, Foundress Vice-Chairman respectively. Under the governance of AICTE & Anna University, this College has 11 branches of study with 8 UG Programmes and 3 PG Programmes
Image Affected with styles
VEL TECH HIGH TECH
Dr.RANGARAJAN Dr.SAKUNTHALA
ENGINEERING COLLEGE
Download Image


Internal CSS
File | C:/Windows/system32/config/systemprofile/Desktop/IP/Ex.No2/InternalCSS.html
Example For Internal CSS
Vel Tech High Tech Dr.Rangarajan Dr.Sakunthala Engineering College was established in the year 2002 under R.S.Trust, founded by the Philanthropic couple: Col. Prof. Vel. Dr. R. Rangarajan, Founder President & Chairman and Dr. Sagunthala Rangarajan, Foundress Vice-Chairman respectively. Under the governance of AICTE & Anna University, this College has 11 branches of study with 8 UG Programmes and 3 PG Programmes
Image Affected with styles
VEL TECH HIGH TECH
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ENGINEERING COLLEGE
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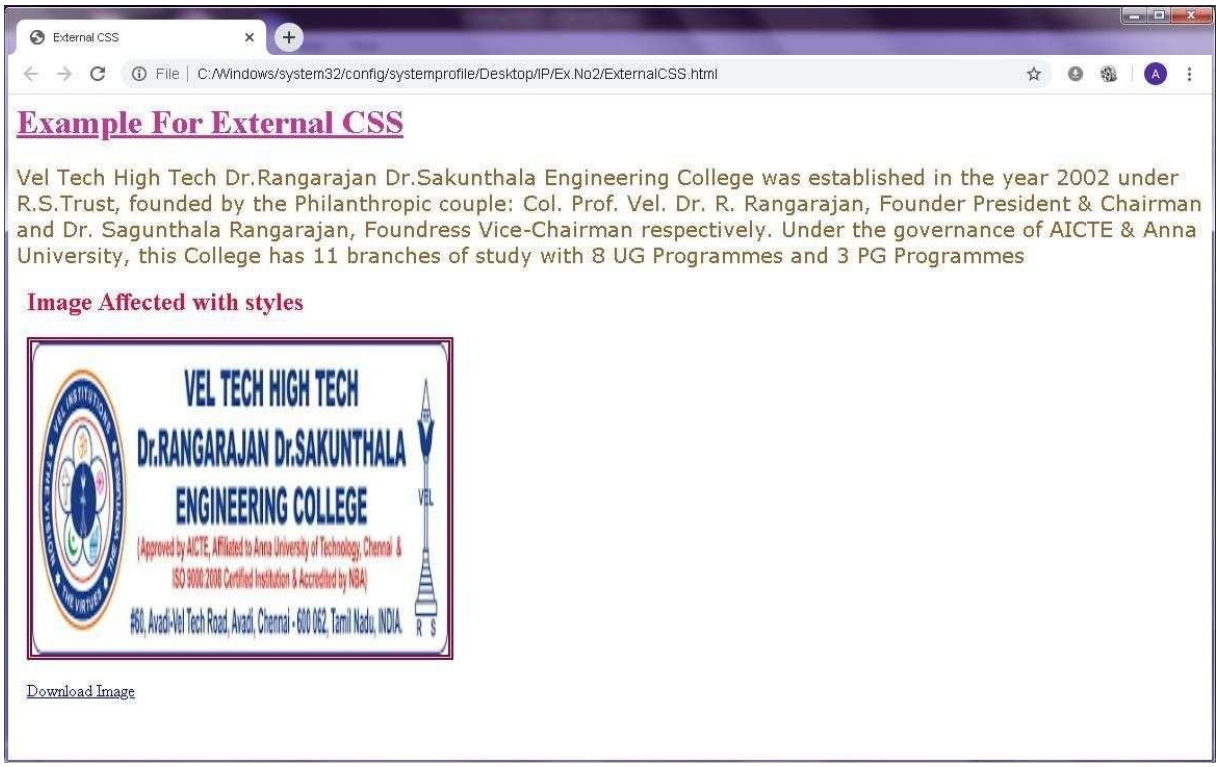


NAME : YUVARAJ G REG NO : 113023104181

RESULT:

Ex.No: 3	FORM VALIDATION USING JAVASCRIPT
DATE:	

AIM:

ALGORITHM:

Login page

Step1: Create Web page using HTML form that contains the fields such as Credit Card Number, amount and a validate button.
Step2: Set the URL of the server as the value of form.(Set registration html as the link when user clicks registration)Redirect to profile.html if username and password is valid
Step3: Run the HTML program.
Step4: Submit the login data to the server.

Profile page

Step1: Create Web page using HTML form that contains the Text fields such as User Name, Password, Login and Cancel buttons.
Step2: Set the URL of the server as the value of form.
Step3: Run the HTML program.
Step4: Submit the data to the server
Step5: Server validates the data and on successful validation it prints a message that payment done, otherwise prints data is invalid.

Form validation

Step1: Create Web page using HTML from that contains the Text fields such as Name, Email, Pin code, a Name of the country Dropdown button and a submit buttons.
Step3: Run the HTML program.
Step4: On Submit the data gets Validated.

PROGRAM:

Loginpage.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Login Page</title>
</head>
<body>
   <h1 style="font-family: Comic Sans MS; font-size: 20pt; color: red; text-align: center;">
      Login Page
   </h1>
   <form name="login">
      Username:
      <input type="text" name="userid" />
      Password:
      <input type="password" name="pswrd" />
      <br><br>
      <input type="button" onclick="check(this.form)" value="Login" />
      <input type="reset" value="Cancel" />
      <br><br>
      <a href="RegForm.html">Register</a>
   </form>
   <script>
      function check(form) {
         if (form.userid.value == "test" && form.pswrd.value == "pass") {
            window.open('profile.html');
         } else {
            alert("Enter valid Username and Password");
         }
      }
   </script>
</body>
</html>
```

Profile.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Profile Page</title>
   <script>
      function validateCard() {
         var numdigits = document.myForm.card.value.length;
         if (numdigits < 9 || isNaN(document.myForm.card.value)) {
            alert("Card number should be greater than 9 digits");
```

```
            return false;
        } else {
            if (
                document.myForm.amt.value != "" &&
                !isNaN(document.myForm.amt.value)
            ) {
                alert("Payment done");
                return true;
            } else {
                alert("Enter the amount");
                return false;
            }
        }
    }
  </script>
</head>
<body>
  <h1>Welcome User</h1>
  <p>You can make payment here</p>
  <form name="myForm" onsubmit="return validateCard();">
    <table>
        <tr>
            <td align="right">Credit Card</td>
            <td><input type="text" name="card" /></td>
        </tr>
        <tr>
            <td align="right">Enter Amount</td>
            <td><input type="text" name="amt" /></td>
        </tr>
        <tr>
            <td></td>
            <td><input type="submit" value="Validate" /></td>
        </tr>
    </table>
  </form>
</body>
</html>
```

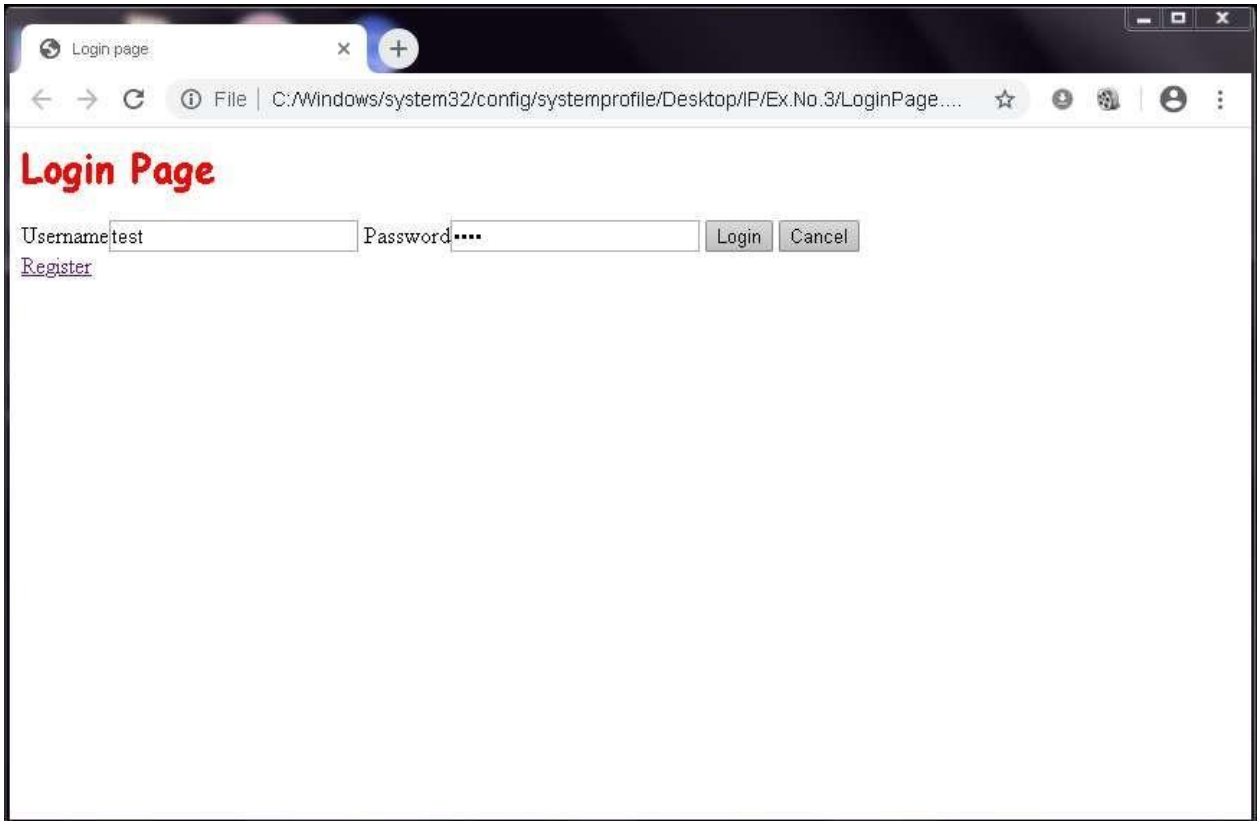
RegForm.html

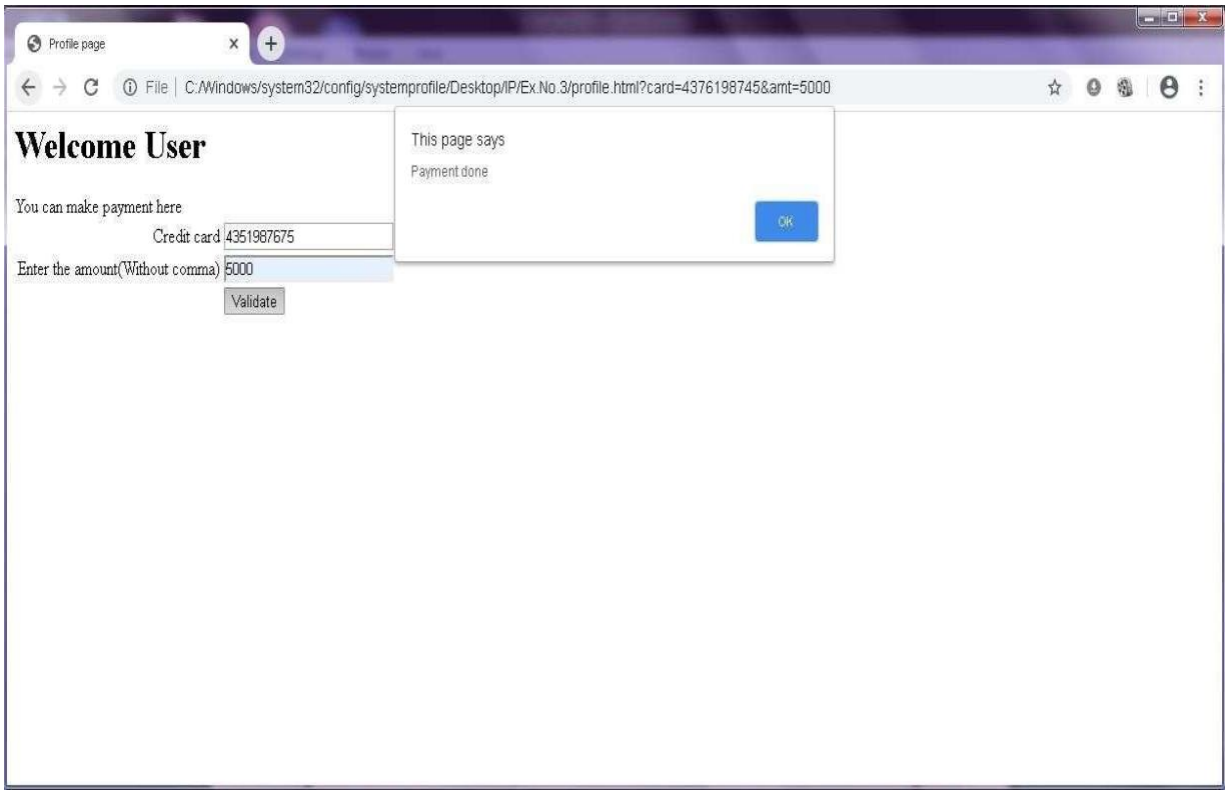
```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>

  <script>
    function validate() {
        if (document.myForm.Name.value == "") {
```

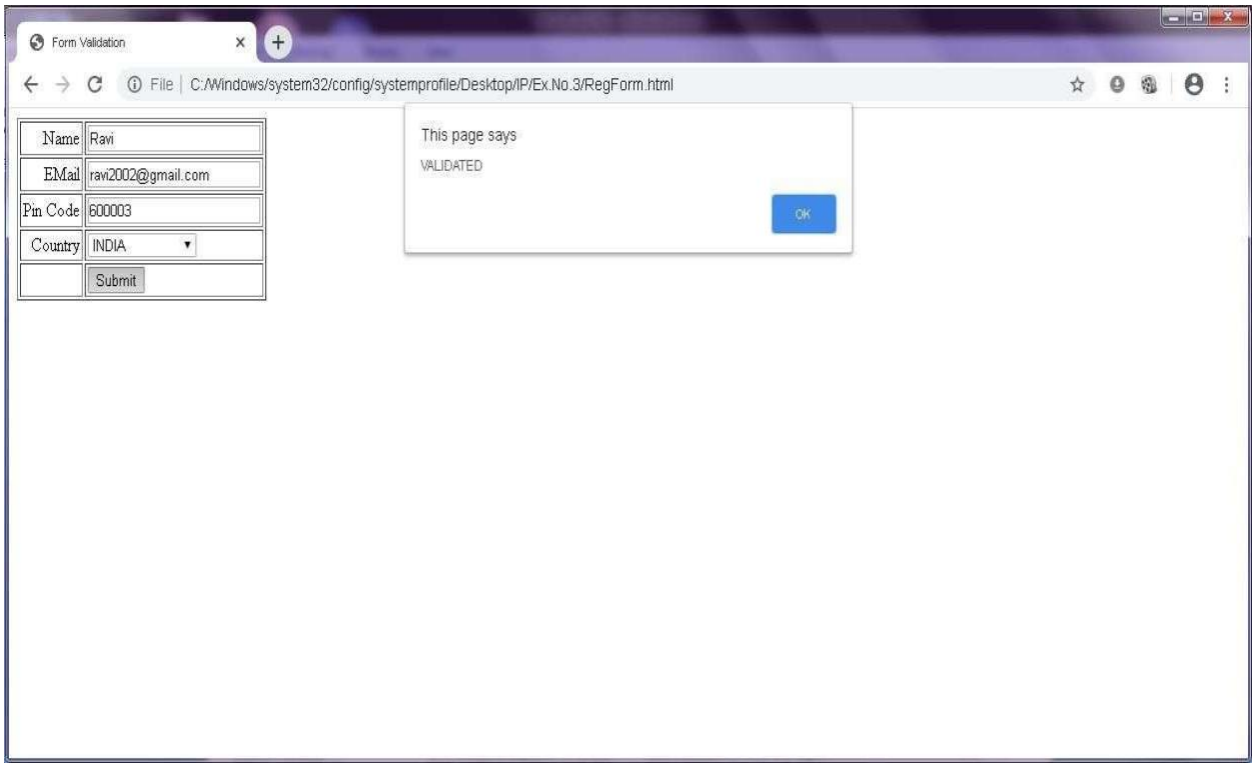
```
            alert("Please provide your name!");
            document.myForm.Name.focus();
            return false;
        }

        if (document.myForm.EMail.value == "") {
            alert("Please provide your Email!");
            document.myForm.EMail.focus();
            return false;
        } else {
            var uemail = document.myForm.EMail.value;
            var mailformat = /^\w+([\.-]?\w+)*@\w+([\.-]?\w+)*(\.\w{2,3})+$/;

            if (!uemail.match(mailformat)) {
                alert("You have entered an invalid email address!");
                document.myForm.EMail.focus();
                return false;
            }
        }

        if (
            document.myForm.Zip.value == "" ||
            isNaN(document.myForm.Zip.value) ||
            document.myForm.Zip.value.length > 6
        ) {
            alert("Please provide a Pin Code lesser than 7 characters.");
            document.myForm.Zip.focus();
            return false;
        }

        if (document.myForm.Country.value == "-1") {
            alert("Please provide your country!");
            return false;
        }

        alert("VALIDATED");
        return true;
    }
  </script>
</head>

<body>

  <form name="myForm" onsubmit="return validate();">
    <table cellspacing="2" cellpadding="2" border="1">

        <tr>
            <td align="right">Name</td>
```

```
            <td><input type="text" name="Name" /></td>
        </tr>

        <tr>
            <td align="right">Email</td>
            <td><input type="text" name="EMail" /></td>
        </tr>

        <tr>
            <td align="right">Pin Code</td>
            <td><input type="text" name="Zip" /></td>
        </tr>

        <tr>
            <td align="right">Country</td>
            <td>
                <select name="Country">
                    <option value="-1" selected>[choose yours]</option>
                    <option value="1">USA</option>
                    <option value="2">UK</option>
                    <option value="3">INDIA</option>
                </select>
            </td>
        </tr>

        <tr>
            <td></td>
            <td><input type="submit" value="Submit" /></td>
        </tr>

    </table>
  </form>

</body>
</html>
```

OUTPUT:


Login page
File | C:/Windows/system32/config/systemprofile/Desktop/IP/Ex.No.3/LoginPage....
Login Page
Username test
Password ••••
Login
Cancel
Register


Profile page
File | C:/Windows/system32/config/systemprofile/Desktop/IP/Ex.No.3/profile.html?card=4376198745&amt=5000
Welcome User
You can make payment here
Credit card 4351987675
Enter the amount(Without comma) 5000
Validate
This page says
Payment done
OK


Form Validation
File | C:/Windows/system32/config/systemprofile/Desktop/IP/Ex.No.3/RegForm.html
Name
Ravi
EMail
ravi2002@gmail.com
Pin Code
600003
Country
INDIA
Submit
This page says
VALIDATED
OK

RESULT:

Ex.No: 4(a)	INVOKING SERVLETS FROM HTML
DATE:	

AIM:

ALGORITHM:

Client.html:

Step1: Create a web page using HTML form that contains the fields such as label,text and one submit button.
Step2: Set the URL of the server as the value of form's action attribute.
Step3: Run the HTML program.
Step4: Submit the form data to the server.

Server.java:

Step1: Define the class server that extends the property of the class GenericServlet.
Step2: Handle the request from the client by using the method service() of GenericServlet class.
Step3: Get the parameter names from the HTML form by using the method getParameterNames().
Step4: Get the parameter values from the HTML forms by using the method getParameter().
Step5: Send the response to the client by using the method of PrintWriter class.

PROGRAM:

Client.html

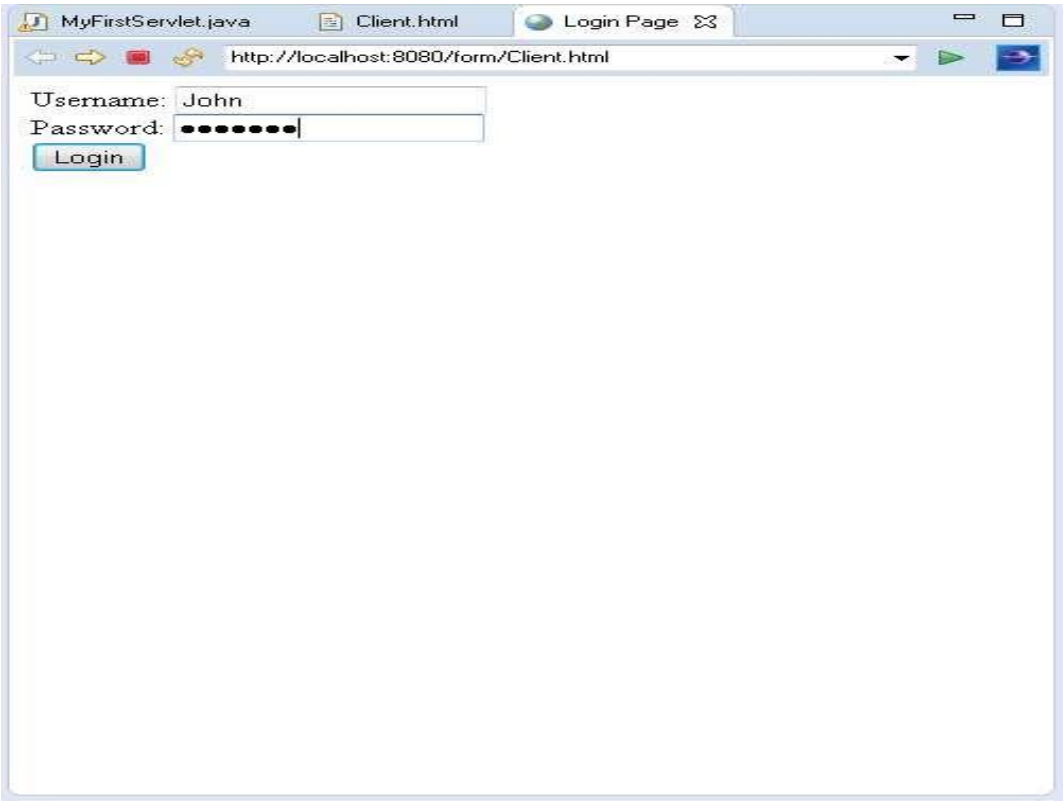
```
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Login Page</title>
</head>
<body>
<form name="loginForm" method="post"
   action="MyFirstServlet"> Username: <input type="text"
   name="username"/> <br/> Password: <input
   type="password" name="password"/> <br/>
   <input type="submit" value="Login" />
</form>
</body>
</html>
```

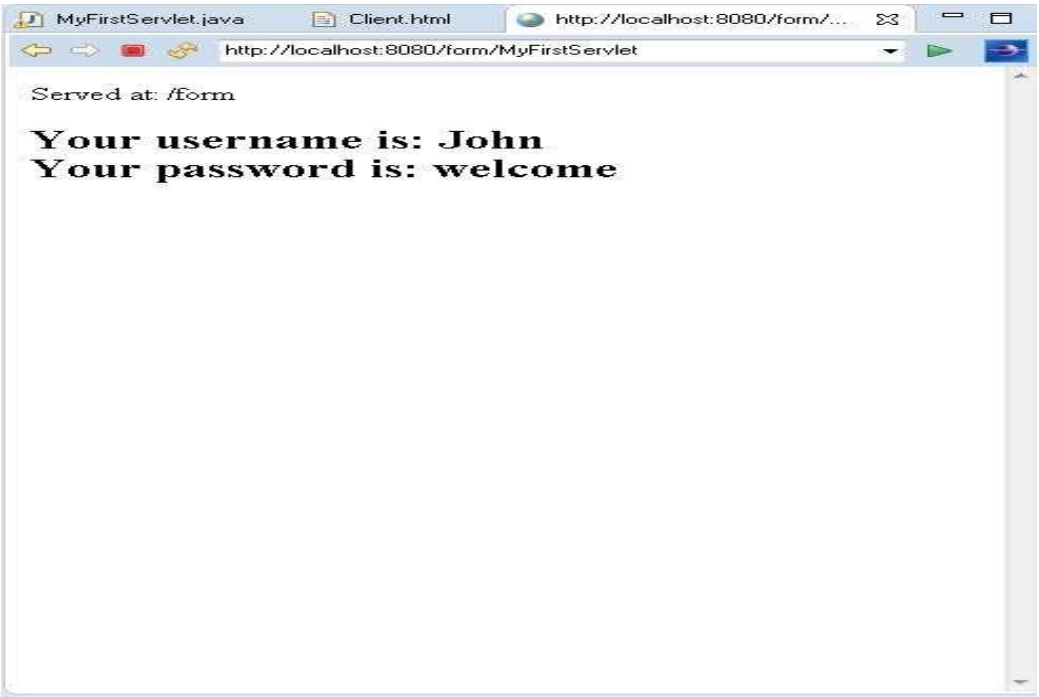
Server.java:

```
import
java.io.IOException;
import
java.io.PrintWriter;
import javax.servlet.ServletException;
import
javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import
javax.servlet.http.HttpServletRequest;
import
javax.servlet.http.HttpServletResponse;

@WebServlet("/MyFirstServlet")
public class MyFirstServlet extends HttpServlet
{
        public MyFirstServlet()
   {

         super();
        System.out.println("Iam created");
   }
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException
 {
      response.getWriter().append("Served at:
      ").append(request.getContextPath()); String username =
```

```
        request.getParameter("username");
        String password = request.getParameter("password");

                System.out.println("username: " +
                username);
                System.out.println("password: " +
                password);

            PrintWriter writer = response.getWriter();
            String htmlRespone = "<html>";
                htmlRespone += "<h2>Your username is: " + username +
                "<br/>"; htmlRespone += "Your password is: " + password +
                "</h2>"; htmlRespone += "</html>";
            writer.println(htmlRespone);
        }
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException
    {
                    doGet(request, response);
    }
}
```

OUTPUT:


MyFirstServlet.java
Client.html
Login Page
http://localhost:8080/form/Client.html
Username: John
Password:
Login


MyFirstServlet.java
Client.html
http://localhost:8080/form/...
http://localhost:8080/form/MyFirstServlet
Served at: /form
Your username is: John
Your password is: welcome

RESULT:

Ex.No: 4b (i)	SESSION TRACKNG USING HIDDEN FORM FIELDS
DATE:	

AIM:

ALGORITHM:

Step 1 : Create a html file which contains user information.

Step 2 : The first servlet will receive these values.

Step 3 : First servlet stores them in the hidden form fields.

Step 4 : The second servlet will display the user information with greeting message.

PROGRAM:

HTML CODE:

```
<html>
<head>
<title>Session Tracking Using Hidden Form Field</title>
</head>
<body>
<form action="Servlet1">
Enter Name:<input type="text" name="username"/><br/>
Enter city:<input type="text" name="usercity"/><br/>
<input type="submit" value="submit"/>
</form>
</body>
</html>
```

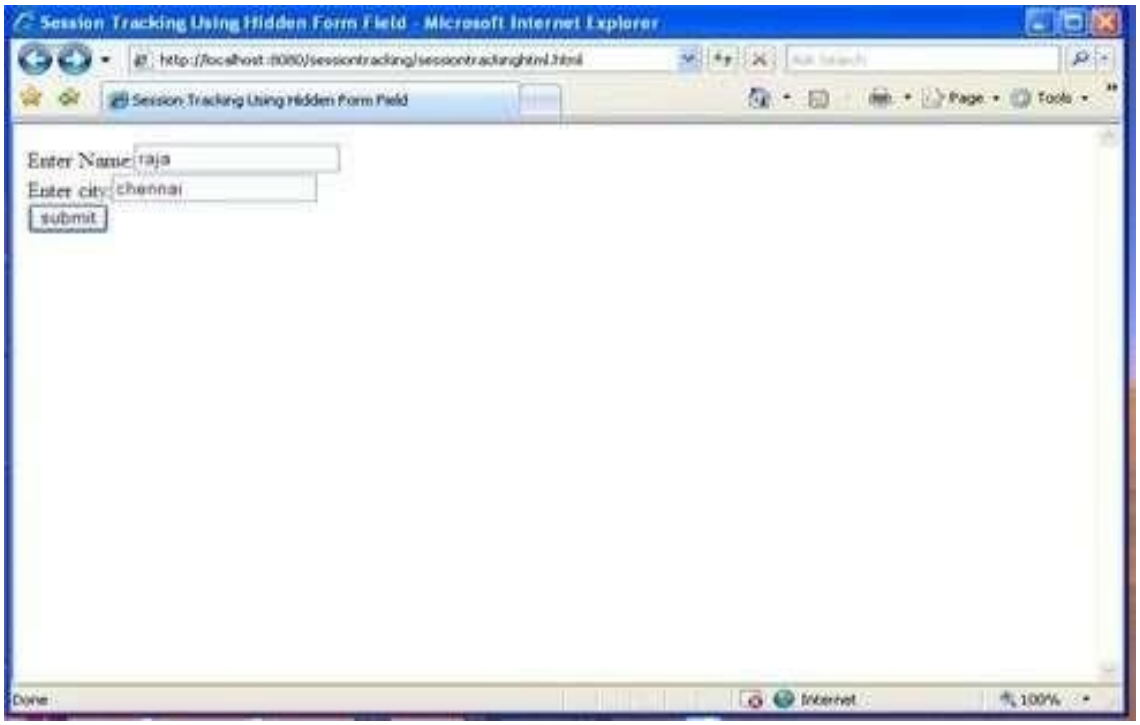
Servlet1:

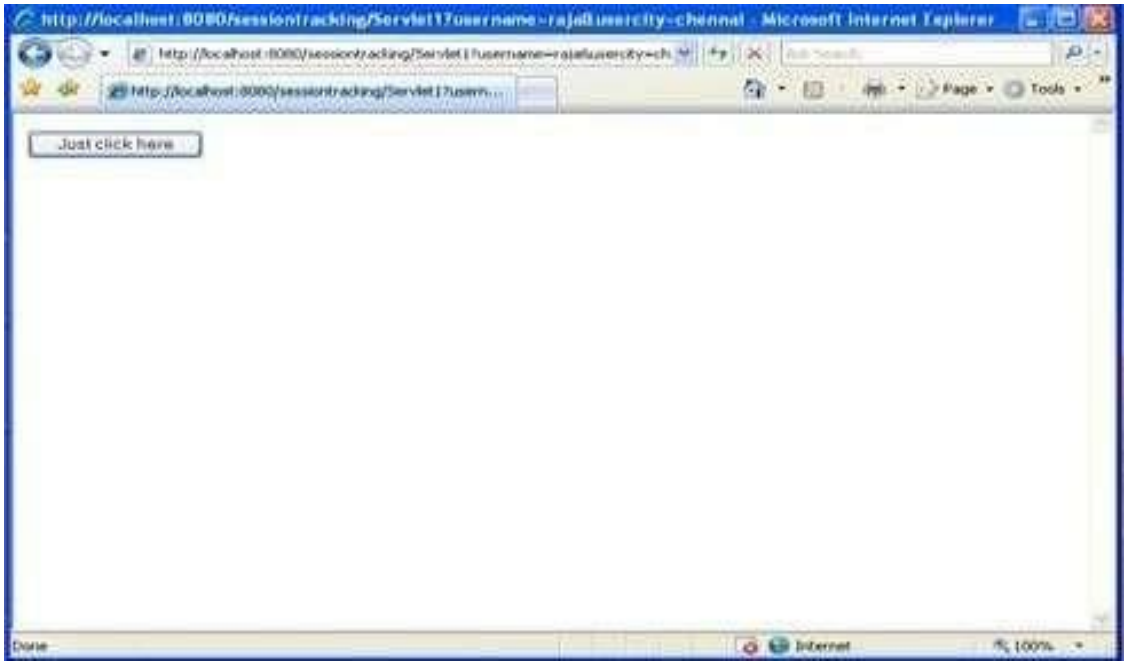
```
import java.io.*; import
javax.servlet.*;
import javax.servlet.http.*;
public class Servlet1 extends HttpServlet
 {
  public void doGet(HttpServletRequest request,HttpServletResponse response)
    {
    try
     {
       response.setContentType("text/html");
       PrintWriter out = response.getWriter(); String
       n=request.getParameter("username"); String
       c=request.getParameter("usercity");
       out.print("<form action=\"Servlet2\">");
       out.print("<input type='hidden' name='uname' value="+n+"'>");
       out.print("<input type='submit' value='Just click here'>");
       out.print("</form>");
       out.close();
     }
    catch(Exception e)
     {
       System.out.println(e);
     }
  }
 }
```

Servlet2:

```
import java.io.*; import
javax.servlet.*;
import javax.servlet.http.*;
public class Servlet2 extends HttpServlet
 {
  public void doGet(HttpServletRequest request,HttpServletResponse response)
   {
    try
     {
       response.setContentType("text/html");
       PrintWriter out = response.getWriter();
       String n=request.getParameter("uname");
       out.println("Hello"+n);
       String c=request.getParameter("usercity");
       out.print("you are from"+c);
       out.close();
     }
    catch(Exception e)
     {
       System.out.println(e);
     }
  }
 }
```

OUTPUT:


Session Tracking Using Hidden Form Field - Microsoft Internet Explorer
Session Tracking Using Hidden Form Field
Enter Name:
raja
Enter city:
chennai
submit
Done
Internet
100%


Just click here
Done
Internet
100%

RESULT:

Ex.No: 4b (ii)	SESSION TRACKING A HIT COUNT
DATE:	

AIM:

ALGORITHM:

Step1: Servlet program to keep track of user visiting the page.

Step2: The count is incremented by one when user visits.

Step3: The output displays the greeting message.

Step4: The number of previous access is also displayed.

PROGRAM:

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/HCounterServlet")
public class HCounterServlet extends HttpServlet {
   public HCounterServlet() {
      super();
   }

   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      HttpSession session = request.getSession(true);

      Integer count = (Integer) session.getAttribute("count");
      String head;

      if (session.isNew() || count == null) {
         count = 1;
         head = "Welcome New User. This is your first visit.";
      } else {
         count = count + 1;
         head = "This is your visit #: " + count;
      }

      session.setAttribute("count", count);
      out.println("<html>");
      out.println("<body bgcolor=\"#FDF5E6\">");
      out.println("<h2 align=\"center\">" + head + "</h2>");
      out.println("</body>");
      out.println("</html>");
   }

   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {

      doGet(request, response);
   }
}
```

OUTPUT:

HCounterServlet.java
http://localhost:8080/count/HCounterServlet
http://localhost:8080/count/HCounterServlet
Served at: /count
Welcome New user.This is your first visit

HCounterServlet.java
http://localhost:8080/count/HCounterServlet
http://localhost:8080/count/HCounterServlet
Served at: /count
This is your :2visit

RESULT:

Ex.No: 5	THREE-TIER APPLICATIONS USING SERVLETS
DATE:	

AIM:

.

ALGORITHM:

Client:

- In index.html on the client side declare the contents that you like to transfer to the server using html form and input type tags.
- Create a submit button and close all the included tags.

Server:

- Import all necessary packages
- Define a class that extends servlet
- In the doPost() method, do the following:
  i) Set the content type of the response to "text/html"
  ii) Create a writer to the response
  iii) Get a paratmeter from the request
  iv) If its value is equal to right answer then add 5 to mark variable
  v) Similarly repeat step for all parameters
  vi) Create a new jsp to read the values from DB and send the result in an html format.

Student Mark List Database:

- Import necessary to java packages and javax packages and classes
- Create a class that extends HttpServlet and implements ServletException
- and IOException
- In the doGet() method, do the following:
  i) Create a PrintWriter object
  ii) Open a connection with the data source name
  iii) Write a sql query and execute to get the resultset
  iv) Display the resultset information in html for

PROGRAM:

index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Online Examination</title>
   <script>
      function validation(form) {
         if (form.Seat_no.value.length === 0) {
            alert("Please fill up the Seat Number");
            form.Seat_no.focus();
            return false;
         }
         if (form.Name.value.length === 0) {
            alert("Please fill up the Name");
            form.Name.focus();
            return false;
         }
         return true;
      }
   </script>
</head>

<body bgcolor="lightgreen">
   <center><h2>Online Examination</h2></center>
   <form action="OEValidator" method="post" name="entry" onsubmit="return
validation(this)">
      <input type="hidden" name="action" value="list">
      <hr>
      <table align="center">
         <tr>
            <td>Seat Number :</td>
            <td><input type="text" name="Seat_no"></td>
         </tr>

         <tr>
            <td>Name :</td>
            <td><input type="text" name="Name" size="50"></td>
         </tr>
      </table>
      <hr>
```

```
        <center>
            <b>Total Marks: 5 × 5 = 25</b>    
            <b>Time: 1 Min.</b>
        </center>

        <br>

        <b>1. Earth is round</b><br>
        <input type="radio" name="group1" value="True"> True
        <input type="radio" name="group1" value="False"> False
        <br><br>

        <b>2. Satellites revolve in space</b><br>
        <input type="radio" name="group2" value="True"> True
        <input type="radio" name="group2" value="False"> False
        <br><br>

        <b>3. You can go to moon via space shuttle</b><br>
        <input type="radio" name="group3" value="True"> True
        <input type="radio" name="group3" value="False"> False
        <br><br>

        <b>4. There is zero gravity in space</b><br>
        <input type="radio" name="group4" value="True"> True
        <input type="radio" name="group4" value="False"> False
        <br><br>

        <b>5. Kalpana Chawla is first Indian woman to go to space</b><br>
        <input type="radio" name="group5" value="True"> True
        <input type="radio" name="group5" value="False"> False
        <br><br>

        <hr>

        <center>
            <input type="submit" value="Submit">
            <input type="reset" value="Clear">
        </center>

    </form>

</body>
</html>
```

JSP Page :

```
<%@ page import="java.sql.Connection" %>
<%@ page import="java.sql.DriverManager" %>
<%@ page import="java.sql.ResultSet" %>
<%@ page import="java.sql.Statement" %>

<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
   pageEncoding="ISO-8859-1"%>

<!DOCTYPE html>
<html>
<head>
   <meta charset="ISO-8859-1">
   <title>Student Results</title>
</head>

<body>

<%
   Connection con = null;
   Statement statement = null;
   ResultSet resultSet = null;

   try {
      Class.forName("net.ucanaccess.jdbc.UcanaccessDriver");

      String filename = "E:/OELabExercise/OEDB.accdb";
      String database = "jdbc:ucanaccess://" + filename;

      con = DriverManager.getConnection(database, "", "");
      statement = con.createStatement();

      resultSet = statement.executeQuery("SELECT * FROM STUDENTTABLE");
%>

<table border="1" align="center" style="text-align: center">
   <thead>
      <tr>
         <th>S.No</th>
         <th>Name</th>
         <th>Marks</th>
      </tr>
   </thead>
```

```
    <tbody>
        <%
            while (resultSet.next()) {
        %>
        <tr>
            <td><%= resultSet.getString(1) %></td>
            <td><%= resultSet.getString(2) %></td>
            <td><%= resultSet.getString(3) %></td>
        </tr>
        <%
            }
        %>
    </tbody>
</table>

<%
    } catch (Exception e) {
        e.printStackTrace();
    }
%>

</body>
</html>
```

OUTPUT:

http://localhost:8080/OnlineExamination/index.html

Online Examination

Seat Number : 1232

Name : Jack

Total Marks:5*5=25 **Time: 1 Min.**

1 . Earth is round
True False

2. Satelites revolve in space.
True False

3. You can go to moon via space shuttle
True False

4There is zero gravity in space.
True False

5. Kapana Chawla is first indian woman to go to space
True False

Submit Clear

JSP

http://localhost:8080/OnlineExamination/resultDisplay.jsp

S.No	NAME	Marks
gh	gggg	15
123435	Joe	15
5565	5	null
1232	Jack	20

RESULT:

Ex.No: 6	INSTALLING TOMCAT AND CONVERTING STATIC WEB PAGES TO DYNAMIC WEB PAGES
DATE :	

AIM:

ALGORITHM:

Step1: We will create a html form for entering the user name,password and card ID.

Step2: From the above HTML form, the servlet program is invoked in which the validity of the user name,password and card id is checked.if it is a valid user then the welcome message will be displayed otherwise the “invalid user” message will be displayed. In this servlet we set the cookies in which the current user name is stored.

Step3: Compile the above servlet Login servlet.java We must store the user information such as user name,password and card id in the init-param using initParams argument of the @webServlet annotation.

Step4: On successful login , the information from the cookie is checked and shopping cart page for corresponding user can be displayed.

Step5: Compile the above servlet LoginSuccess.java

Step6: Start tomcat web server.Open the web browser and display the login form created in step1.

PROGRAM

Login.jsp:

```
<!DOCTYPE html>
<html>
<head>
   <title>Login Page</title>
</head>

<body>

   <form action="LoginServlet" method="post">
      Enter Username:
      <input type="text" name="user">
      <br><br>

      Enter Password:
      <input type="password" name="password">
      <br><br>

      Enter Card ID:
      <input type="text" name="cardID">
      <br><br>

      <input type="submit" value="Login">
   </form>

</body>
</html>
```

LoginServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebInitParam;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
@WebServlet(
    urlPatterns = {"/LoginServlet"},
    initParams = {
        @WebInitParam(name = "usernames", value = "user1,user2,user3"),
        @WebInitParam(name = "passwords", value = "pwd1,pwd2,pwd3"),
        @WebInitParam(name = "cardIDs", value = "111,222,333")
    }
)
public class LoginServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    public LoginServlet() {
        super();
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        try {
            String usr = request.getParameter("user");
            String pwd = request.getParameter("password");
            String card = request.getParameter("cardID");

            boolean invalid = true;

            String[] userID = getInitParameter("usernames").split(",");
            String[] password = getInitParameter("passwords").split(",");
            String[] cardids = getInitParameter("cardIDs").split(",");

            for (int i = 0; i < userID.length; i++) {

                if (userID[i].equals(usr) &&
                    password[i].equals(pwd) &&
                    cardids[i].equals(card)) {

                    invalid = false;

                    Cookie myCookie = new Cookie("currentUser", usr);
                    myCookie.setMaxAge(60 * 60); // 1 hour
                    response.addCookie(myCookie);
```

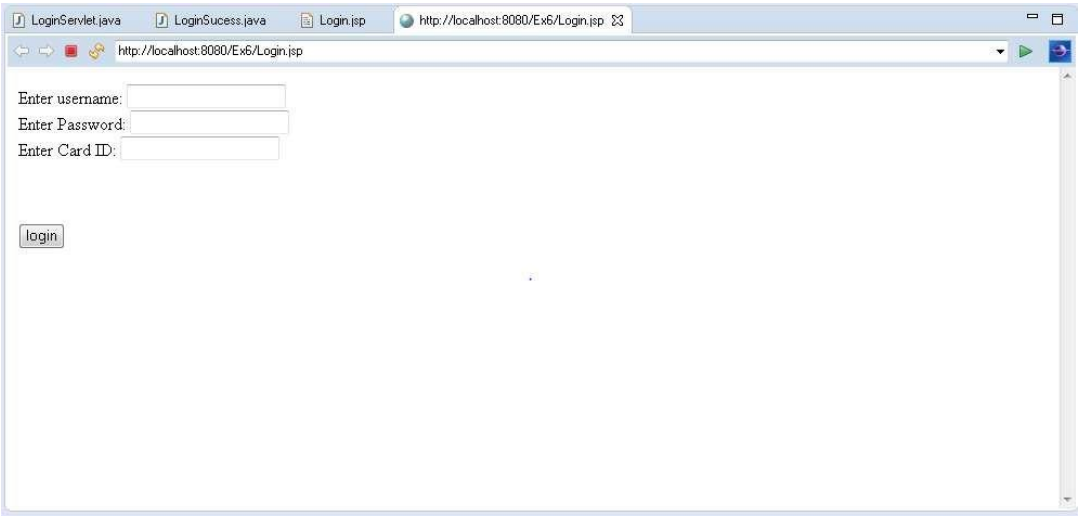
```
                response.sendRedirect("LoginSucess");
                return;
            }
        }

        if (invalid) {
            out.println("<h4>Invalid user, please try again</h4>");
            out.println("<a href='login.html'>Login Again</a>");
        }

    } finally {
        out.close();
    }
}

protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    doGet(request, response);
}
}
```

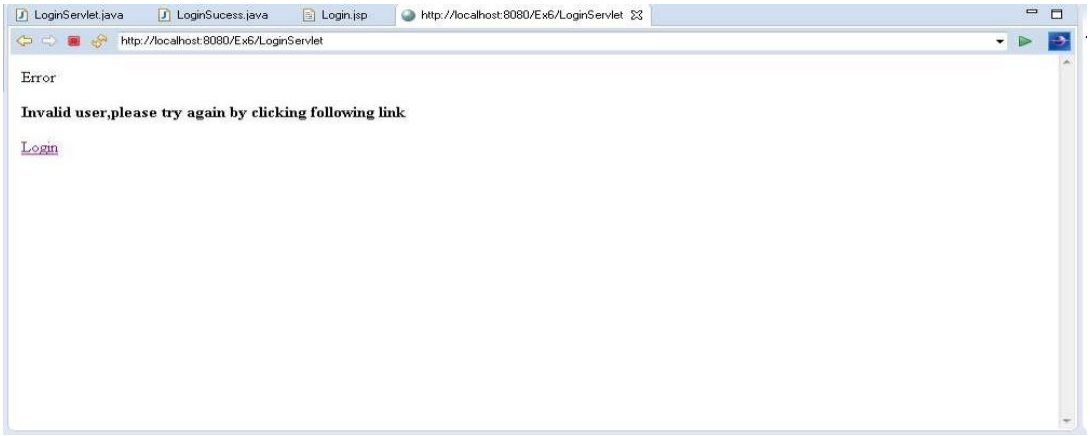
LoginSucess.java

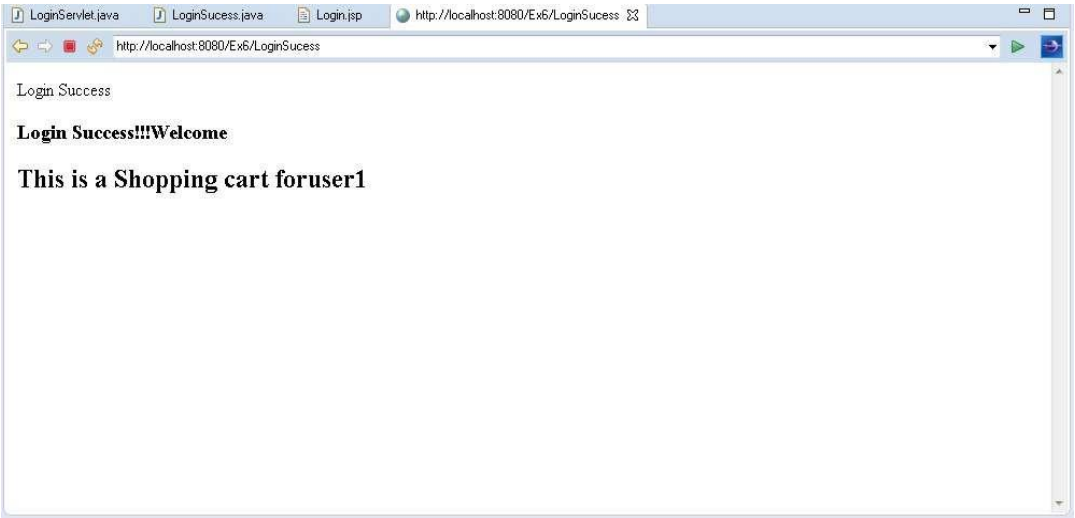
```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/LoginSucess")
public class LoginSucess extends HttpServlet {

    private static final long serialVersionUID = 1L;

    public LoginSucess() {
        super();
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
```

```
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String userName = null;
        Cookie[] cookies = request.getCookies();

        if (cookies != null) {
            for (Cookie cookie : cookies) {
                if (cookie.getName().equals("currentUser")) {
                    userName = cookie.getValue();
                }
            }
        }

        out.println("<h2>Login Success!!!</h2>");

        if (userName != null) {
            out.println("<h3>Welcome " + userName + "</h3>");
            out.println("<h4>This is a Shopping Cart for " + userName + "</h4>");
        } else {
            out.println("<h4>User not found in cookies</h4>");
        }

        out.close();
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        doGet(request, response);
    }
}
```

OUTPUT:


LoginServlet.java
LoginSucess.java
Login.jsp
http://localhost:8080/Ex6/Login.jsp
http://localhost:8080/Ex6/Login.jsp
Enter username:
Enter Password:
Enter Card ID:
login


LoginServlet.java
LoginSucess.java
Login.jsp
http://localhost:8080/Ex6/LoginServlet
http://localhost:8080/Ex6/LoginServlet
Error
Invalid user,please try again by clicking following link
Login


LoginServlet.java
LoginSucess.java
Login.jsp
http://localhost:8080/Ex6/LoginSucess
http://localhost:8080/Ex6/LoginSucess
Login Success
Login Success!!!Welcome
This is a Shopping cart foruser1

RESULT:

Ex.No: 7	CONVERTING STATIC WEB PAGES TO DYNAMIC WEB PAGES (USER INFORMATION AND BOOKS INFORMATION)
DATE:	

AIM:

ALGORITHM:

Step 1: Create your own directory under tomcat/webapps (e.g. tr1)
Step 2: Copy the html files in tr1
Step 3: Copy the jsp files also into tr1
Step 4: Start tomcat give the following commandCatalina.bat runAt install-dir/bin
Step 5: At I.E give url as http://localhost:8081/tr1/main.html

PROGRAM:

Main.html:

```
<html>
<body bgcolor="pink">
   <br><br><br><br><br><br>
   <h1 align="center"><u>ONLINE BOOK STORAGE</u></h1>
   <br><br><br>
   <h2 align="center">
      <pre>
<b>
Welcome to online book storage.
Press LOGIN if you are having id
Otherwise press REGISTRATION
</b>
      </pre>
   </h2>
   <br><br>
   <div align="center">
      <a href="/tr/login.html">LOGIN</a> |
      <a href="/tr/reg.html">REGISTRATION</a>
   </div>
</body>
</html>
```

Login.html

```
<html>
<body bgcolor="pink">
   <br><br><br>
   <form name="myform" method="post" action="/tr1/login.jsp">
      <div align="center">
         <pre>
LOGIN ID  : <input type="text" name="id">
PASSWORD  : <input type="password" name="pwd">
         </pre>
      </div>
      <br><br>
      <div align="center">
         <input type="submit" value="OK">
             
         <input type="reset" value="CLEAR">
      </div>
   </form>
</body>
</html>
```

Reg.html

```
<html>
<body bgcolor="pink">
   <br><br>
   <form name="myform" method="post" action="/tr1/reg.jsp">
      <div align="center">
         <pre>
NAME           : <input type="text" name="name">
ADDRESS        : <input type="text" name="addr">
CONTACT NUMBER  : <input type="text" name="phno">
LOGIN ID       : <input type="text" name="id">
PASSWORD       : <input type="password" name="pwd">
         </pre>
      </div>
      <br><br>
      <div align="center">
         <input type="submit" value="OK">
             
         <input type="reset" value="CLEAR">
      </div>
   </form>
</body>
</html>
```

Profile.html

```
<html>
<body bgcolor="pink">
   <br><br>
   <form method="post" action="/tr1/profile.jsp">
      <div align="center">
         <pre>
LOGIN ID : <input type="text" name="id">
         </pre>
      </div>
      <br><br>
      <div align="center">
         <input type="submit" value="OK">
             
         <input type="reset" value="CLEAR">
      </div>
   </form>
</body>
</html>
```

Catalog.html

```
<html>
<body bgcolor="pink">
   <br><br><br>
   <form method="post" action="/tr1/catalog.jsp">
      <div align="center">
         <pre>
BOOK TITLE : <input type="text" name="title">
         </pre>
      </div>
      <br><br>
      <div align="center">
         <input type="submit" value="OK">
             
         <input type="reset" value="CLEAR">
      </div>
   </form>
</body>
</html>
```

Order.html

```
<html>
<body bgcolor="pink">
   <br><br><br>
   <form method="post" action="/tr1/order.jsp">
      <div align="center">
         <pre>
LOGIN ID          : <input type="text" name="id">
PASSWORD          : <input type="password" name="pwd">
TITLE             : <input type="text" name="title">
NO. OF BOOKS      : <input type="text" name="no">
DATE              : <input type="text" name="date">
CREDIT CARD NO    : <input type="password" name="cno">
         </pre>
      </div>
      <br><br>
      <div align="center">
         <input type="submit" value="OK">
             
         <input type="reset" value="CLEAR">
      </div>
   </form>
</body>
</html>
```

Login.jsp

```
<%@ page import="java.sql.*" %>
<%
out.println("<html><body bgcolor='pink'>");
String id = request.getParameter("id");
String pwd = request.getParameter("pwd");
DriverManager.registerDriver(new oracle.jdbc.driver.OracleDriver());
Connection con = DriverManager.getConnection(
    "jdbc:oracle:thin:@localhost:1521:orcl",
    "scott",
    "tiger"
);
Statement stmt = con.createStatement();
String sql = "SELECT id, password FROM login WHERE id='" + id + "' AND password='" +
pwd + "'";
ResultSet rs = stmt.executeQuery(sql);
if (!rs.next()) {
    out.println("SORRY INVALID ID<br><br>");
    out.println("<a href='/tr1/login.html'>Retry</a>");
} else {
    out.println("VALID LOGIN<br><br>");
    out.println("<ul>");
    out.println("<li><a href='profile.html'>USER PROFILE</a></li>");
    out.println("<li><a href='catalog.html'>BOOKS CATALOG</a></li>");
    out.println("<li><a href='order.html'>ORDER</a></li>");
    out.println("</ul>");
}
out.println("</body></html>");
%>
```

Reg.jsp

```
<%@ page import="java.sql.*" %>
<%
out.println("<html><body bgcolor='pink'>");
String name = request.getParameter("name");
String addr = request.getParameter("addr");
String phno = request.getParameter("phno");
String id = request.getParameter("id");
String pwd = request.getParameter("pwd");
DriverManager.registerDriver(new oracle.jdbc.driver.OracleDriver());
Connection con = DriverManager.getConnection(
    "jdbc:oracle:thin:@localhost:1521:orcl",
    "scott",
    "tiger"
);
```

```
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery("SELECT id FROM login");
boolean exists = false;
while (rs.next()) {
    if (id.equals(rs.getString(1))) {
        exists = true;
        break;
    }
}
if (exists) {
    out.println("LOGIN ID EXISTS<br><br>");
    out.println("<a href='/tr1/reg.html'>Retry</a>");
} else {
    stmt.executeUpdate(
        "INSERT INTO login VALUES('" + name + "','" + addr + "','" + phno + "','" + id + "','"
+ pwd + "')"
    );
    out.println("REGISTERED SUCCESSFULLY<br><br>");
    out.println("<a href='/tr1/login.html'>Login</a>");
}
out.println("</body></html>");
%>
```

Profile.jsp

```
<%@ page import="java.sql.*" %>
<%
out.println("<html><body bgcolor='pink'>");
String id = request.getParameter("id");
DriverManager.registerDriver(new oracle.jdbc.driver.OracleDriver());
Connection con = DriverManager.getConnection(
    "jdbc:oracle:thin:@localhost:1521:orcl",
    "scott",
    "tiger"
);
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery("SELECT * FROM login WHERE id='" + id + "'");
if (rs.next()) {
    out.println("<center>");
    out.println("NAME: " + rs.getString(1) + "<br>");
    out.println("ADDRESS: " + rs.getString(2) + "<br>");
    out.println("PHONE: " + rs.getString(3) + "<br>");
    out.println("</center>");
} else {
    out.println("INVALID ID<br>");
}
out.println("</body></html>");
%>
```

Catalog.jsp

```
<%@ page import="java.sql.*" %>
<%
out.println("<html><body bgcolor='pink'>");
String title = request.getParameter("title");
DriverManager.registerDriver(new oracle.jdbc.driver.OracleDriver());
Connection con = DriverManager.getConnection(
    "jdbc:oracle:thin:@localhost:1521:orcl",
    "scott",
    "tiger"
);
Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery("SELECT * FROM book WHERE title='" + title + "'");
if (rs.next()) {
    out.println("<center>");
    out.println("TITLE: " + rs.getString(1) + "<br>");
    out.println("AUTHOR: " + rs.getString(2) + "<br>");
    out.println("VERSION: " + rs.getString(3) + "<br>");
    out.println("PUBLISHER: " + rs.getString(4) + "<br>");
    out.println("COST: " + rs.getString(5) + "<br>");
    out.println("</center>");
} else {
    out.println("BOOK NOT FOUND<br>");
}
out.println("</body></html>");
%>
```

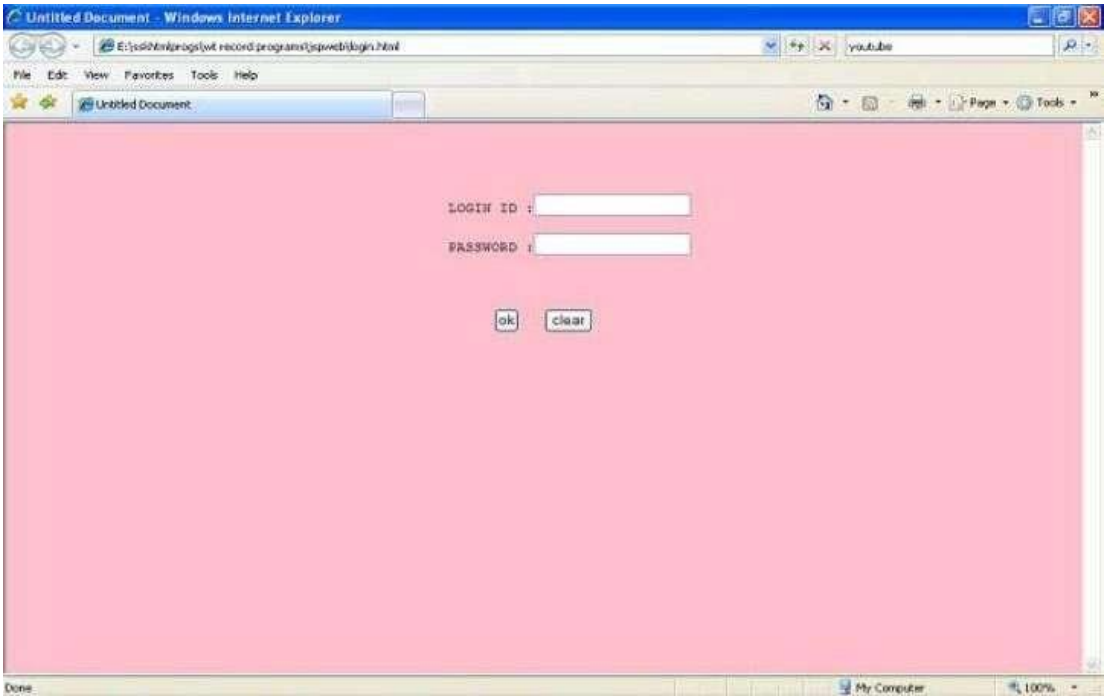
Order.jsp

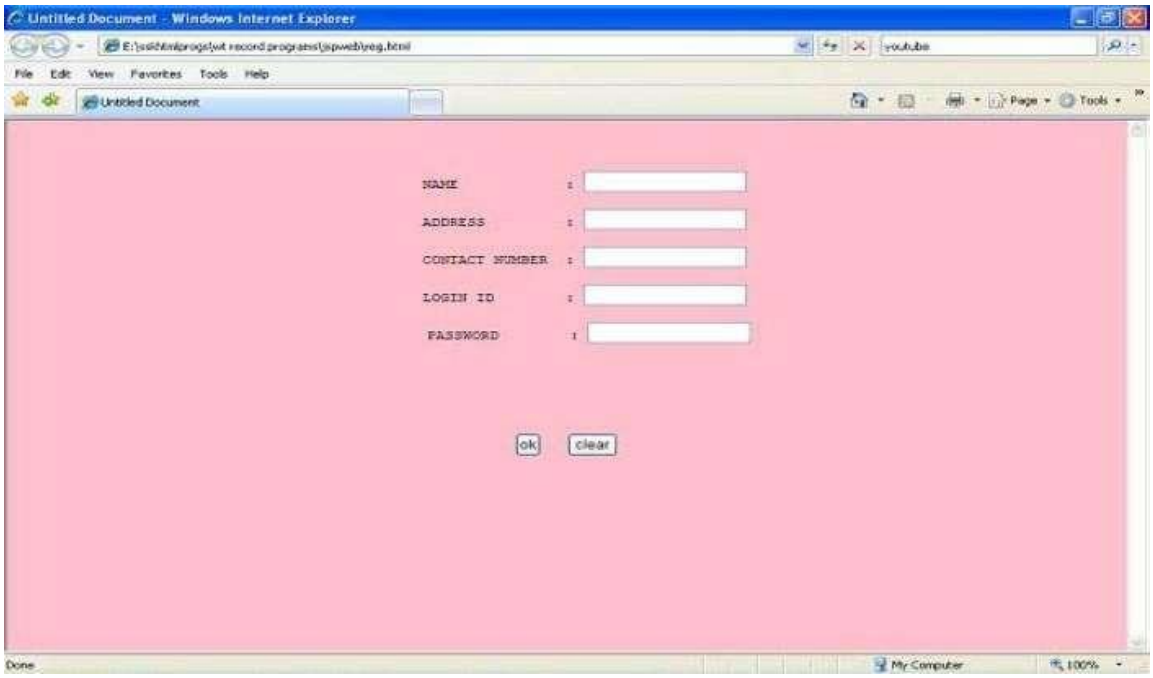
```
<%@ page import="java.sql.*" %>

<%
out.println("<html><body bgcolor='pink'>");

String id = request.getParameter("id");
String pwd = request.getParameter("pwd");
String title = request.getParameter("title");
int count = Integer.parseInt(request.getParameter("no"));
String date = request.getParameter("date");
String cno = request.getParameter("cno");

DriverManager.registerDriver(new oracle.jdbc.driver.OracleDriver());
```

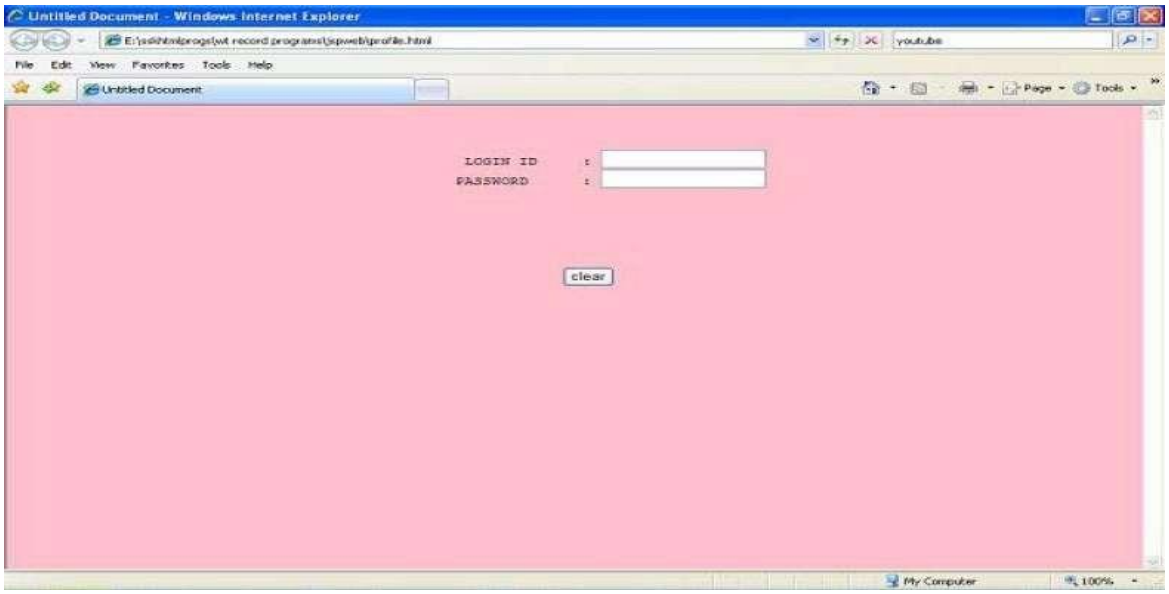
```
Connection con = DriverManager.getConnection(
    "jdbc:oracle:thin:@localhost:1521:orcl",
    "scott",
    "tiger"
);

Statement stmt = con.createStatement();
ResultSet rs = stmt.executeQuery("SELECT * FROM login WHERE id='" + id + "' AND
password='" + pwd + "'");

if (!rs.next()) {
    out.println("INVALID LOGIN<br>");
} else {
    ResultSet rs1 = stmt.executeQuery("SELECT cost FROM book WHERE title='" + title +
"'");

    if (rs1.next()) {
        int cost = Integer.parseInt(rs1.getString(1));
        int amount = cost * count;

        out.println("AMOUNT: " + amount + "<br>");

        stmt.executeUpdate(
            "INSERT INTO details VALUES('" + id + "','" + title + "','" + amount + "','" + date +
"','" + cno + "')"
        );

        out.println("ORDER PLACED");
    } else {
        out.println("INVALID BOOK");
    }
}

out.println("</body></html>");
%>
```

OUTPUT:

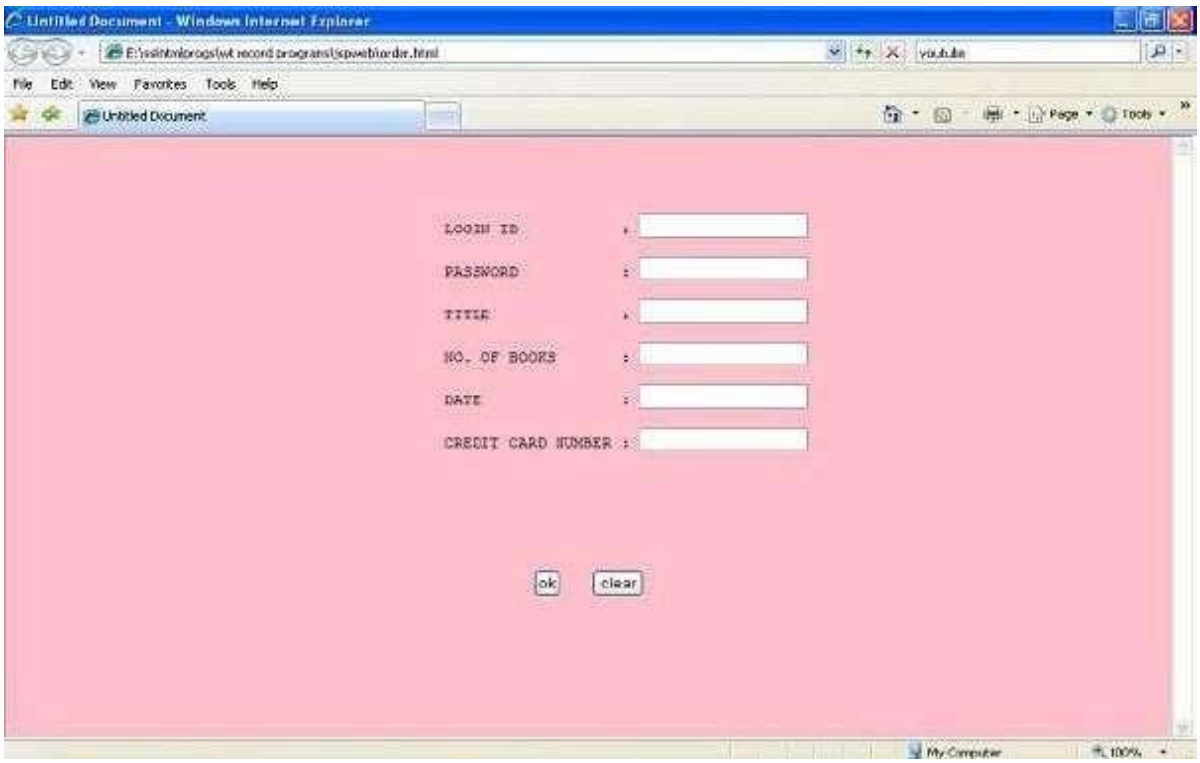
Untitled Document - Windows Internet Explorer
File Edit View Favorites Tools Help
Untitled Document
ONLINE BOOK STORAGE
Welcome to online book storage.
Press LOGIN if you are having id
Otherwise press REGISTRATION
LOGIN
REGISTRATION
My Computer
100%


Untitled Document - Windows Internet Explorer
File Edit View Favorites Tools Help
Untitled Document
LOGIN ID :
PASSWORD :
ok
clear
Done
My Computer
100%


Untitled Document - Windows Internet Explorer
File Edit View Favorites Tools Help
Untitled Document
NAME :
ADDRESS :
CONTACT NUMBER :
LOGIN ID :
PASSWORD :
ok
clear
Done
My Computer
100%


Untitled Document - Windows Internet Explorer
File Edit View Favorites Tools Help
Untitled Document
LOGIN ID :
PASSWORD :
clear
My Computer
100%

Untitled Document - Windows Internet Explorer
File Edit View Favorites Tools Help
Untitled Document
BOOK TITLE :
ok
clear
My Computer
100%


Untitled Document - Windows Internet Explorer
File Edit View Favorites Tools Help
Untitled Document
LOGIN ID :
PASSWORD :
TITLE :
NO. OF BOOKS :
DATE :
CREDIT CARD NUMBER :
ok
clear
My Computer
100%

RESULT:

Ex.No:8	RETRIEVING USER INFORMATION FROM XML DOCUMENT
DATE:	

AIM:

ALGORITHM:

Step 1: Save Students information in the XML file on the specific location.
Step 2: Create and establish the connection between html file and XML file.
Step 3: Get the user ID as input
Step 4: Display the student's information

PROGRAM:

StudentDetails.xml :

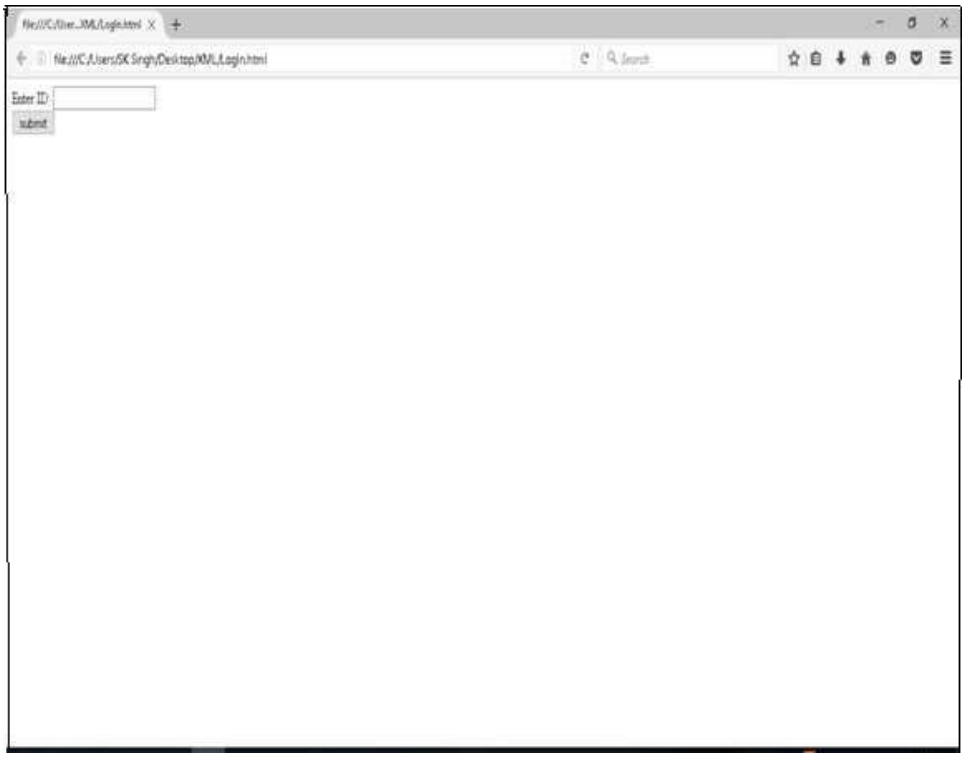
```
<?xml version="1.0" encoding="UTF-8"?>
<Student>
   <PersonDetails>
      <id>101</id>
      <name>Anand</name>
      <city>Madurai</city>
      <Branch>CSE</Branch>
      <Year>I</Year>
   </PersonDetails>

   <PersonDetails>
      <id>102</id>
      <name>Anu</name>
      <city>Konam</city>
      <Branch>CSE</Branch>
      <Year>II</Year>
   </PersonDetails>

   <PersonDetails>
      <id>103</id>
      <name>Archana</name>
      <city>Madurai</city>
      <Branch>CSE</Branch>
      <Year>I</Year>
   </PersonDetails>

   <PersonDetails>
      <id>104</id>
      <name>Monica</name>
      <city>Nellai</city>
      <Branch>CSE</Branch>
      <Year>III</Year>
   </PersonDetails>
</Student>
```

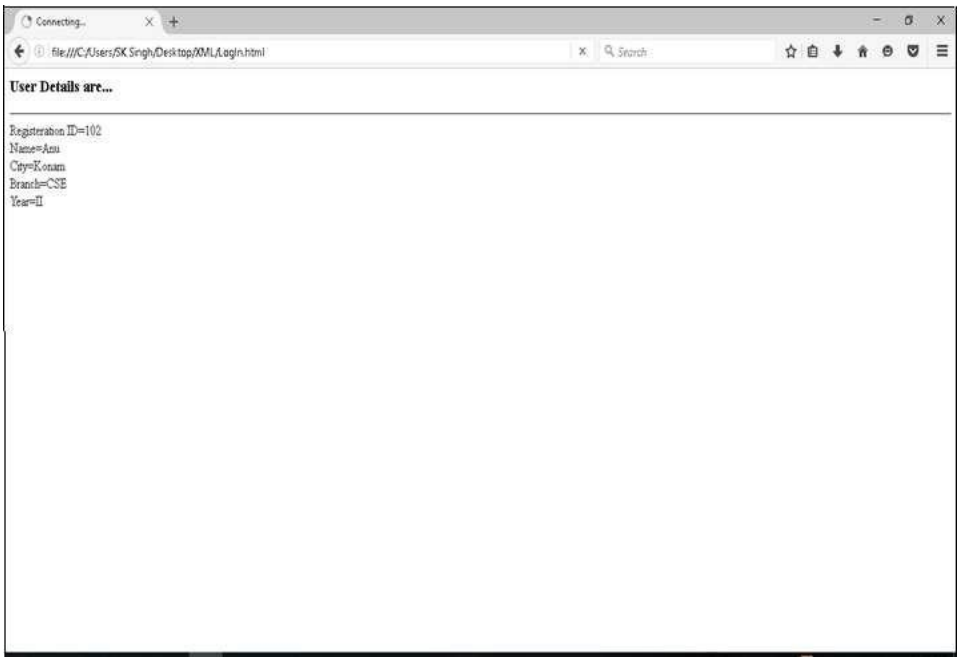
LogIn.html

```
<!DOCTYPE html>
<html>
<head>
</head>
```

```
<body>
   <script type="text/javascript">
      function Display() {
         if (window.XMLHttpRequest) {
            xmlhttp = new XMLHttpRequest();
         }
         xmlhttp.open("GET", "UserInfo.xml", false);
         xmlhttp.send();
         xmlDoc = xmlhttp.responseXML;

         var x = xmlDoc.getElementsByTagName("PersonDetails");
         var key_id = document.getElementById("key").value;

         for (i = 0; i < x.length; i++) {
            if (key_id.match(
               x[i].getElementsByTagName("id")[0].childNodes[0].nodeValue
            )) {
               j = i;
            }
         }
         document.write("<h3>User Details are...</h3> <hr> Registeration ID=");
         document.write(x[j].getElementsByTagName("id")[0].childNodes[0].nodeValue);
         document.write("</br> Name=");
document.write(x[j].getElementsByTagName("name")[0].childNodes[0].nodeValue);
         document.write("</br> City=");
         document.write(x[j].getElementsByTagName("city")[0].childNodes[0].nodeValue);
         document.write("</br> Branch=");

 document.write(x[j].getElementsByTagName("Branch")[0].childNodes[0].nodeValue);

         document.write("</br> Year=");

 document.write(x[j].getElementsByTagName("Year")[0].childNodes[0].nodeValue);

         document.write("</br> ");
      }
   </script>

   <form name="myform">
      Enter ID:
      <input type="text" id="key" /><br />
      <input type="button" value="submit" onclick="Display()" />
   </form>
</body>
</html>
```

OUTPUT:


file:///C:/Users/SK Singh/Desktop/XML/Login.html
Enter ID:
submit


Connecting...
file:///C:/Users/SK Singh/Desktop/XML/Login.html
User Details are...
Registeration ID=102
Name=Anu
City=Konam
Branch=CSE
Year=II

RESULT:

Ex.No:9	VALIDATING AND STORING A FORM DATA INTO DATABASE USING PHP
DATE:	

AIM:

ALGORITHM:

Step 1: Install XAMPP web server.
Step 2: Write a program in a notepad and save it as filename.php.
Step 3: After completion of the installation, we can use the XAMPP Control Panel to start/ stop all servers.
Step 4: Start Mysql and Apache servers.
Step 5: Copy filename.php to htdocs (C:/Program Files/XAMPP/htdocs)
Step 6: Now to run our code, open localhost/filename.php then it gets executed.

PROGRAM:

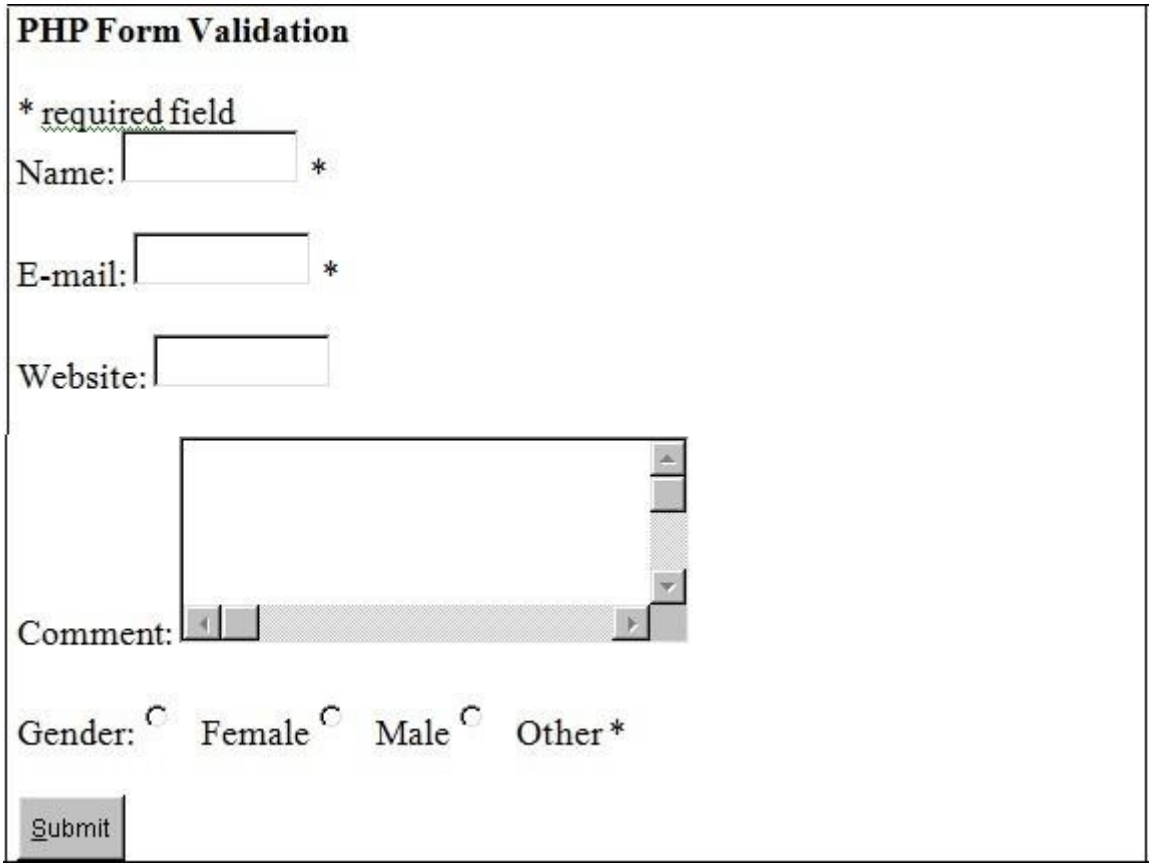
Validate.php

```
<!DOCTYPE HTML>
<html>
<head>
   <style>
      .error {
         color: #FF0000;
      }
   </style>
</head>

<body>

<?php
// define variables and set to empty values
$nameErr = $emailErr = $genderErr = $websiteErr = "";
$name = $email = $gender = $comment = $website = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {

   if (empty($_POST["name"])) {
      $nameErr = "Name is required";
   } else {
      $name = test_input($_POST["name"]);

      // check if name only contains letters and whitespace
      if (!preg_match("/^[a-zA-Z ]*$/", $name)) {
         $nameErr = "Only letters and white space allowed";
      }
   }

   if (empty($_POST["email"])) {
      $emailErr = "Email is required";
   } else {
      $email = test_input($_POST["email"]);

      // check if e-mail address is well-formed
      if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
         $emailErr = "Invalid email format";
      }
   }
```

```
    if (empty($_POST["website"])) {
        $website = "";
    } else {
        $website = test_input($_POST["website"]);

        // check if URL address syntax is valid
        if (!preg_match(
            "/\b(?:(?:https?|ftp):\/\/|www\.)[-a-z0-9+&@#\/%?=~_|!:,.;]*[-a-z0-9+&@#\/%=~_|]/i",
            $website
        )) {
            $websiteErr = "Invalid URL";
        }
    }

    if (empty($_POST["comment"])) {
        $comment = "";
    } else {
        $comment = test_input($_POST["comment"]);
    }

    if (empty($_POST["gender"])) {
        $genderErr = "Gender is required";
    } else {
        $gender = test_input($_POST["gender"]);
    }
}

function test_input($data) {
    $data = trim($data);
    $data = stripslashes($data);
    $data = htmlspecialchars($data);
    return $data;
}
?>

<h2>PHP Form Validation</h2>
<p><span class="error">* required field</span></p>

<form method="post" action="<?php echo htmlspecialchars($_SERVER["PHP_SELF"]);
?>">
    Name:
    <input type="text" name="name">
    <span class="error">* <?php echo $nameErr; ?></span>
    <br><br>
```

```
    E-mail:
    <input type="text" name="email">
    <span class="error">* <?php echo $emailErr; ?></span>
    <br><br>

    Website:
    <input type="text" name="website">
    <span class="error"><?php echo $websiteErr; ?></span>
    <br><br>

    Comment:
    <textarea name="comment" rows="5" cols="40"></textarea>
    <br><br>

    Gender:
    <input type="radio" name="gender" value="female">Female
    <input type="radio" name="gender" value="male">Male
    <input type="radio" name="gender" value="other">Other
    <span class="error">* <?php echo $genderErr; ?></span>
    <br><br>

    <input type="submit" name="submit" value="Submit">
</form>

<?php
echo "<h2>Your Input:</h2>";
echo $name;
echo "<br>";
echo $email;
echo "<br>";
echo $website;
echo "<br>";
echo $comment;
echo "<br>";
echo $gender;
?>

</body>
</html>
```

OUTPUT:



Your Input: sangamithra
sangamithra@gmail.com
www.google.com
good
female

insert.php

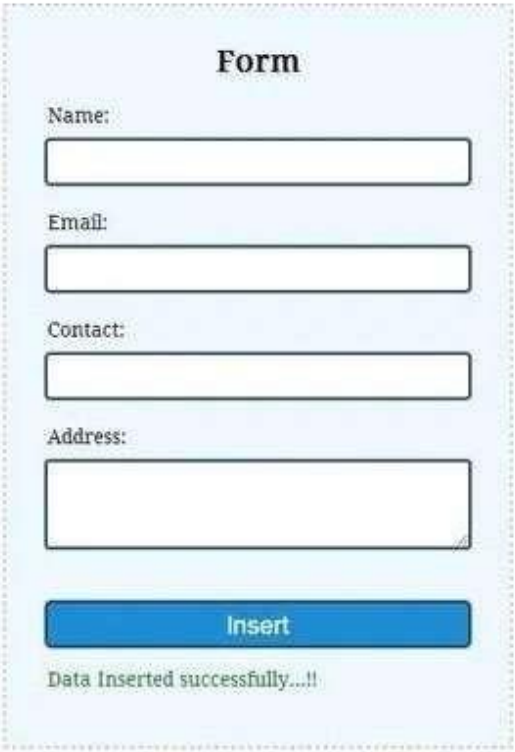
```
<!DOCTYPE html>
<html>
<head>
    <title>PHP insertion</title>
    <link href="css/insert.css" rel="stylesheet">
</head>
<body>
    <div class="maindiv">
        <!-- HTML Form -->
        <div class="form_div">
            <div class="title">
                <h2>Insert Data In Database Using PHP.</h2>
            </div>
            <form action="insert.php" method="post">
                <!-- Method can be set as POST for hiding values in URL -->
                <h2>Form</h2>
                <label>Name:</label>
                <input class="input" name="name" type="text" value="">
                <label>Email:</label>
                <input class="input" name="email" type="text" value="">
                <label>Contact:</label>
                <input class="input" name="contact" type="text" value="">
                <label>Address:</label>
                <textarea cols="25" name="address" rows="5"></textarea>
                <br>
                <input class="submit" name="submit" type="submit" value="Insert">
            </form>
        </div>
    </div>
</body>
</html>
```

conn.php

```php
<?php
$connection = mysql_connect("localhost", "root", "");
$db = mysql_select_db("colleges", $connection);

if (isset($_POST['submit'])) {

    $name = $_POST['name'];
    $email = $_POST['email'];
    $contact = $_POST['contact'];
    $address = $_POST['address'];

    if ($name != '' || $email != '') {

        // Insert Query of SQL
        $query = mysql_query(
            "INSERT INTO students
            (student_name, student_email, student_contact, student_address)
            VALUES ('$name', '$email', '$contact', '$address')"
        );

        echo "<br/><br/><span>Data Inserted successfully...!!</span>";
    } else {
        echo "<p>Insertion Failed <br/> Some Fields are Blank. !!</p>";
    }
}

mysql_close($connection); // Closing Connection with Server
?>
```

OUTPUT:

Insert Data In
Database
Using PHP.


Form
Name:
Email:
Contact:
Address:
Insert
Data Inserted successfully...!!

RESULT:

Ex.No:10	A SURVEY USING WEB SERVICE FOR GETTING A FEEDBACK ON ANY CONSUMER PRODUCT
DATE :	

AIM:

ALGORITHM:

Step 1: Open the home page.
Step 2: Enter the login ID and type the comments then submit.
Step 3: Retrieve comments with post id
Step 4: Display the comments.

PROGRAM:

Index.php

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <title>jQuery Ajax Comment System - Demo</title>
    <link rel="stylesheet" href="css/style.css">
    <script src="http://ajax.googleapis.com/ajax/libs/jquery/1.10.2/jquery.min.js"></script>
    <script src="js/script.js"></script>
</head>

<body>
    <div class="wrap">
        <h1>Maggy Noodles Comment System</h1>
        <?php
        $query = mysql_query('SELECT * FROM post WHERE post_id = 1');
        $row = mysql_fetch_array($query);
        // retrieve comments with post id
        $comment_query = mysql_query("SELECT * FROM comment WHERE post_id = 1");
        ?>
        <div class="post">
            <h2><?php echo $row['post_title']; ?></h2>
            <p><?php echo $row['post_body']; ?></p>
        </div>
        <div class="comment-block">
            <?php while ($comment = mysql_fetch_array($comment_query)): ?>
                <div class="comment-item">
                    <div class="comment-avatar">
                        <img src="<?php echo avatar($comment['mail']); ?>" alt="avatar">
                    </div>

                    <div class="comment-post">
                        <h3>
                            <?php echo $comment['name']; ?>
                            <span>said.</span>
                        </h3>
                        <p><?php echo $comment['comment']; ?></p>
                    </div>
                </div>
            <?php endwhile; ?>
        </div>
```

```
        <h2>Submit new comment</h2>

        <!-- comment form -->
        <form id="form" method="post">
            <input type="hidden" name="postid" value="<?php echo $row['post_id']; ?>">

            <label>
                <span>Name *</span>
                <input type="text" name="name" id="comment-name" placeholder="Your name
here." required>
            </label>

            <label>
                <span>Email *</span>
                <input type="email" name="mail" id="comment-mail" placeholder="Your mail
here." required>
            </label>

            <label>
                <span>Your comment *</span>
                <textarea name="comment" id="comment" cols="30" rows="10"
                    placeholder="Type your comment here." required></textarea>
            </label>

            <input type="submit" id="submit" value="Submit Comment">
        </form>
    </div>
</body>
</html>
```

Ajax_Comment.php

```
<?php
if (isset($_SERVER['HTTP_X_REQUESTED_WITH'])):

    include('config.php');
    include('function.php');
    dbConnect();

    if (!empty($_POST['name']) &&
        !empty($_POST['mail']) &&
        !empty($_POST['comment']) &&
        !empty($_POST['postid'])) {
```

```
        $name = mysql_real_escape_string($_POST['name']);
        $mail = mysql_real_escape_string($_POST['mail']);
        $comment = mysql_real_escape_string($_POST['comment']);
        $postId = mysql_real_escape_string($_POST['postid']);

        mysql_query("
            INSERT INTO comment (name, mail, comment, post_id)
            VALUES ('{$name}', '{$mail}', '{$comment}', '{$postId}')
        ");
    }
?>

<div class="comment-item">
    <div class="comment-avatar">
        <img src="<?php echo avatar($mail); ?>" alt="avatar">
    </div>

    <div class="comment-post">
        <h3>
            <?php echo $name; ?> <span>said.</span>
        </h3>
        <p><?php echo $comment; ?></p>
    </div>
</div>

<?php endif; ?>
```

Config.php

```
<?php
// DB configuration
define('DB_HOST', 'localhost');
define('DB_USER', 'root');
define('DB_PASS', 'root');
define('DB_NAME', 'dbname');
?>
```

Function.php

```php
<?php
/**
 * Connect to mysql server
 */
function dbConnect($close = true) {
    if (!$close) {
        mysql_close();
        return true;
    }

    $link = mysql_connect(DB_HOST, DB_USER, DB_PASS)
        or die('Could not connect to MySQL DB: ' . mysql_error());

    if (!mysql_select_db(DB_NAME, $link)) {
        return false;
    }

    return $link;
}

/**
 * Gravatar Image
 */
function avatar($mail, $size = 60) {
    $url = "http://www.gravatar.com/avatar/";
    $url .= md5(strtolower(trim($mail)));
    $url .= "?s=" . $size;
    return $url;
}
?>
```

style.css

```css
/* General styling */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font: 12px Arial, Tahoma, Helvetica, sans-serif;
    text-rendering: optimizeLegibility;
}
```

```
.wrap {
    width: 720px;
    margin: 15px auto;
    padding: 15px 20px;
    background: #fff;
    border: 2px solid #DBDBDB;
    border-radius: 5px;
}

a {
    text-decoration: none;
    color: #333;
}

h1 {
    font-family: Georgia, "Times New Roman", Times, serif;
    font-size: 2.8em;
    text-align: center;
    margin: 25px 0;
}

h2 {
    font-size: 1.5em;
    margin: 8px 0;
}

h3 {
    font-size: 1.2em;
    margin: 5px 0;
}

.comment-block {
    margin: 20px 0 20px 20px;
}

.comment-item {
    overflow: hidden;
    width: 500px;
    padding: 10px;
    border: 1px solid #E6E6E6;
    border-radius: 5px;
    margin: 5px;
}
```

```
.comment-avatar {
    width: 60px;
    float: left;
}

.comment-avatar img {
    width: 60px;
    height: 60px;
    border-radius: 5px;
}

.comment-post {
    width: 400px;
    float: left;
    padding: 0 5px 0 10px;
}
```

script.js

```
$(document).ready(function () {
    var form = $('#form');
    var submit = $('#submit');

    form.on('submit', function (e) {
        e.preventDefault();

        $.ajax({
            url: 'ajax_comment.php',
            type: 'POST',
            cache: false,
            data: form.serialize(),

            beforeSend: function () {
                submit.val('Submitting...').attr('disabled', 'disabled');
            },

            success: function (data) {
                var item = $(data).hide().fadeIn(800);
                $('.comment-block').append(item);

                form.trigger('reset');
                submit.val('Submit Comment').removeAttr('disabled');
            }
        });
    });
});
```

OUTPUT:

Maggie Noodles Comment System
Comments.....
henry said...
asdf said...
A comment
hgjd said...
ad said...

RESULT:

Ex.No:11	CREATING A HTML5 VIDEO PLAYER AND GETTING FEEDBACK
DATE:	

AIM:

ALGORITHM:

Step 1 : Create html5 page and save it

Step 2: Open html body tag and start video tag

Step 3 : Inside video tag add the videos source and mime tag

Step 4 : If video not supported display appropriate massage inside tag and close video tag

Step 5 : Open textarea tag with needed rows and columns

Step 6 : Create a submit button to get user feedback

Step 7 : OnClick of submit call a JavaScript function to display an alert that feedback has been received and close the textarea tag

Step 8 : Save the file and open in HTML5 enabled browser

PROGRAM:

video.html

```
<!DOCTYPE html>
<html>
<body>

<video id="playlist" controls="controls">
  <source src="C:\video1.mp4" type="video/mp4">
  <source src="c:\video2.mp4" type="video/ogg">
  Your browser does not support HTML5 video.
</video>
<BR>
Your Feedback :
</BR>
<textarea name="comments" rows="10" cols="80">
</textarea>
</BR>
<input type="submit" value="Submit" onclick="subFed();">

<script type='text/javascript'>
    documentgetElementById('playlist').addEventListener('ended',myHandler,false); function
    myHandler(e)
   {
       if(!e)
      {
      e = window.event;
      }
       alert("Video Finished");
     }
           function subFed()
          {
           alert("Feedback Submitted");
           }
</script>
</body>
</html>
```

OUTPUT:

File | C:/Windows/system32/config/systemprofile/Desktop/html/text.html
0:00
Your Feedback
Submit

RESULT:

Ex.No:12	TO ENCODE AND DECODE A STRING USING JAVA
DATE:	

AIM:

ALGORITHM:

Step 1 : Create a java file and in the main method use a hard coded string to node
Step 2 : Create an object for Base64 encoder
Step 3 : Get bytes of a string and pass it to Base64 object encoding method and string will be encoded
Step 4 : To decode the string pas the encoded string to decode method of Base64 object and string will be decoded
Step 5 : Close the main method

PROGRAM:

StringEncode.java

```
import java.nio.charset.StandardCharsets;
import java.util.Base64;
import java.util.Base64.Decoder;
import java.util.Base64.Encoder;

public class StringEncode
{

        public static void main(String[] args)
 {

                String string = "VELTECH HIGHTECH";

                // Encoding
                Encoder encoder = Base64.getEncoder();
                byte[] data = string.getBytes(StandardCharsets.UTF_8); String
                encodedString = encoder.encodeToString(data);
                System.out.println("Given String to encode is " + string);
                System.out.println("Encoded: " + encodedString);

                // Decoding
                Decoder decoder = Base64.getDecoder();
                byte[] bytes = decoder.decode(encodedString);
                String decodedString = new String(bytes, StandardCharsets.UTF_8);

                System.out.println("Decoded: " + decodedString);

        }

}
```

OUTPUT:

```
Problems  @ Javadoc  Declaration  Console
<terminated> StringEncode [Java Application] C:\Program Files\Java\jre1.8.0_144\bin\javaw.exe (Nov 2, 2020, 5:55:37 PM)
Given String to encode is  VELTECH HIGHTECH
Encoded: VkVNVEVDSCBISUdIVEVDSA==
Decoded: VELTECH HIGHTECH
```

RESULT: